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Analysis of Course and Degree Results in a University Programme: Relationships between choices of courses in final year and class of Honours Degree

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Abstract

School of Mathematics & Statistics

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**Analysis of Course and Degree Results in a University Programme:
Relationships between choices of courses in final year and class of
Honours Degree**

by Panthakan BOONSURIYATHAM

University undergraduate students are often interested in knowing which combinations of courses are most likely to lead to the award of the highest class of degrees. This project aims to find how level 4 course choices, the programme of study and graduation year affect the aggregate score and degree classification. We investigate the relationship using statistical methods, including statistical tests and models. We give some advice about suggested courses leading to the highest degree at last.

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Chapter 1

Introduction to the problem

University undergraduate students are often interested in knowing which combinations of courses are most likely to lead to the award of the highest class of degree.

1.1 Discussion of the context

The class degree awards of universities are the goal of every undergraduate student. The UK has four different degree classifications: first, upper second, lower second, and third class. However, each university in the UK has different criteria for classifications because they use different grading systems.

The University of Glasgow uses a 22-point scale grading system, *Guide to the Code of Assessment - Chapter 2 2022*, (see Figure A.1). From A to G, we can tell the differences within each primary grade. In the school of mathematics and statistics, the university awards degree classifications on the aggregate score of each student as described in Figure A.2. The school calculate these scores from courses each student enrolled in the third and the fourth year, called level 3 and level 4, respectively. For example, if they have an overall aggregate over 17.5, equivalent to grade A, the student will get the first class award. If their overall scores fall between two of the ranges in Figure A.2, school determine using Figure A.3.

Each student in our data enrolls statistical courses. The data will contain the results of every course they have enrolled in level 4. Some pieces of information are also provided, such as the level 3 aggregate score, the overall aggregate score, and the degree classification of each student.

1.2 Aims of the research

This project aims to find how level 4 courses affect the aggregate score and degree classification. If there is such a relationship, we should be able to give some information about which combination of courses will lead to the highest degree.

1.3 Questions of interest

The main research questions of this project are

1. Are there differences in the distribution of course results?
2. Are there differences in the course and degree results between the different programmes of study?
3. Are there any differences in the course and degree results between years of graduation?
4. What is the relationship between the aggregate result from third year statistics courses and the overall aggregate score and class of degree awarded?
5. What is the relationship between the choice of final year statistics courses and the overall aggregate score and class of degree awarded? If there is a relationship, which courses should be chosen in the final year of study to maximize the chances of a first class degree?

1.4 Description of the study and variables involved

Two datasets are given in this project as XLSX files. We will mainly study them using R programming and analyse them statistically. There are 38 variables in this study from those files, namely,

1. Year, an academic year that student graduates. We have four academic years:
 - 2015/16
 - 2016/17
 - 2017/18
 - 2018/19
2. Level, a number showing level of each student when they enrol on courses. It is always 4 in our data.
3. Programme, a description of the degree programme that each student was enrolled on, namely:
 - Single - Single Honours in Statistics
 - Maths & Stats - Combined Honours in Statistics & Mathematics
 - Stats & Other - Combined Honours in Statistics & a non-Maths Subject
 - Erasmus - Statistics as part of an EU Programme
4. Student.Num, a unique number identifying each student

5. 28 statistical courses, a result of each course in the format of the secondary band. The complete list of courses in the data is, alphabetically,

S001	S002	S003	S005	S006	S007
S008	S009	S010	S011	S012	S013
S014	S016	S017	S018	S019	S021
S022	S023	S024	S025	S026	S027
S028	S029	SProj.20Cr	SProj.30Cr		

Every course is worth ten credits, except for SProj.20Cr and SProj.30Cr which are worth 20 and 30 credits, respectively.

6. Credits.Level3, a number showing the total number of credits for the third year results
7. Credits.Level4, a number showing the total number of credits for the fourth year results
8. Level3Aggregate, a number showing the combined course results in the third year ranging from 0 to 22
9. Level4Aggregate, a number showing the combined course results in the fourth year ranging from 0 to 22
10. Overall.Aggregate, a number showing the combined course results from the third and the fourth year ranging from 0 to 22
11. Degree.Classification, a number showing a degree classification on 5 point scale¹, as defined in Figure A.2 and A.3, namely:
 - 1 - the first class degree
 - 2.1 - the upper second class degree
 - 2.2 - the lower second class degree
 - 3 - the third class degree

¹The degree classifications are cut on the overall aggregate score in every enrolled course and subject, while the overall aggregate score in our data is only from courses in statistics. Some students will have their overall score in all subjects higher or lower than in statistics, so the degree classification will differ from what we expect in our data.

Chapter 2

Description of the methods

We describe the statistical methods that have been used in this project

2.1 Student's t-Test

Also known as **two-sample t-test** or **independent sample t-test**, see Macdonald, 2021. It is used to test under the null hypothesis the mean of two populations are equal. The test assumptions are

1. Each sample is observed from a normally distributed population
2. Each sample is observed from a population of the same variance
3. Each sample is independent to each other

The test statistic is

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\left(\frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}\right) (1/n_1 + 1/n_2)}},$$

where $\bar{x}_1$ and $\bar{x}_2$ are sample means, s_1^2 and s_2^2 are sample variance, and n_1 and n_2 are sample sizes. Under the null hypothesis, $t \sim t(n_1 + n_2 - 2)$.

2.2 F Test to Compare Two Variances

It is used to test under the null hypothesis the variances of two populations are equal, see Dalgaard, 2011. The test assumptions are, each sample is observed from a normally distributed population and they are independent. The test statistic is

$$F = \frac{s_1^2}{s_2^2},$$

where s_1^2 and s_2^2 are sample variances, and n_1 and n_2 are sample sizes. Under the null hypothesis, $F \sim F(n_1 - 1, n_2 - 1)$.

2.3 Pearson's Chi-squared Test

It is used to test under the null hypothesis two categorical variables, of m and n categories, are independent, see Macdonald, 2021. The test assumption is each sample is a random sampling from a fixed distribution. The test statistic is

$$X^2 = \sum_{i=1}^m \sum_{j=1}^n \frac{(O_{ij} - E_{ij})^2}{E_{ij}},$$

where O_{ij} is the observed frequency and $E_{ij} = \frac{\sum_{k=1}^m O_{kj} \times \sum_{l=1}^n O_{il}}{\sum_{k=1}^m \sum_{l=1}^n O_{kl}}$ is the expected frequency. Under the null hypothesis, $X^2 \sim \chi^2((m-1)(n-1))$.

2.4 Linear model

The linear model is the tool to predict the value of a continuous response variable from various explanatory variables. It is the most common model and usually in the form $Y = X\beta + \epsilon$. For more details, see James et al., 2017.

2.5 Multinomial and Proportional odd logistic regression

We use multinomial regression when the response variables are categorical. It gives the probability of the response variable being in each category, the probabilities sum up to one in each observation.

In addition, if those categorical variables have an order, we may use a proportional odd logistic regression instead. In this model, the number of equations depends on the number of categories. Each equation is in the form of

$$\log \left(\frac{p_1 + \dots + p_k}{p_{k+1} + \dots + p_n} \right) = \beta_{0,k} + X\beta,$$

where $k \in \{1, \dots, n-1\}$, $0 \leq p_i \leq 1$ is the probability of being the i^{th} category when ordered. For their more details, see Zhang and Neocleous, 2022.

2.6 Brier Score

The Brier score is a function measuring the accuracy of probabilistic prediction, see Wilks, 2010. The score ranges from 0 to 1, the lower score, the better prediction. The score can be calculated by the formula

$$\text{Brier Score} = \frac{1}{n} \sum_{i=1}^n (p_i - o_i)^2,$$

where p_i is the predicted probability, o_i takes the value 0 and 1, if the outcome does not happen and happen, respectively.

Chapter 3

Analysis of the data

Before we analyse the data, we shall prepare them by handling missing data and organising the format.

3.1 Data Preparation

There are two files in our project. After we carefully look, we can see that one file is an extension of another. Hence, this project will read and analyse only one file, which is **MScProj2020MB1.xlsx**. The file contains records of 190 students.

Next, we will handle some missing data, which we will do in the following

1. Course results (every course listed in Section 1.4) are not modified and treated that not enrol.
2. Level 3 and overall aggregate results (Level3Aggregate and Overall.Aggregate, respectively) are not modified. There are 12 and only students who have missing scores in both variables. They have the graduation year 2017/18 and the Erasmus programme in common. So we believe and assume that these Erasmus students do not have degree classification because they do not enrol on enough courses as a university requirement - the missing data have their own meanings.
3. Degree classification (Degree.Classification) will be managed in two cases. There are two groups of students; one is the group of Erasmus students in 2017/18 who also have missing data in level 3 and overall - we will leave this classification as missing and will not be considered. Another one is the rest students shown as "NA" in Figure 3.1 (left). Their classifications will be filled using the values of the overall aggregate results (Overall.Aggregate) by the following cutpoints:

Overall aggregate result x	Degree classification
$17.5 \leq x \leq 22$	First class (1)
$14.5 \leq x < 17.5$	Upper second class (2:1)
$11.5 \leq x < 14.5$	Lower second class (2:2)
$8.5 \leq x < 11.5$	Third class (3)
$0 \leq x < 8.5$	Fail

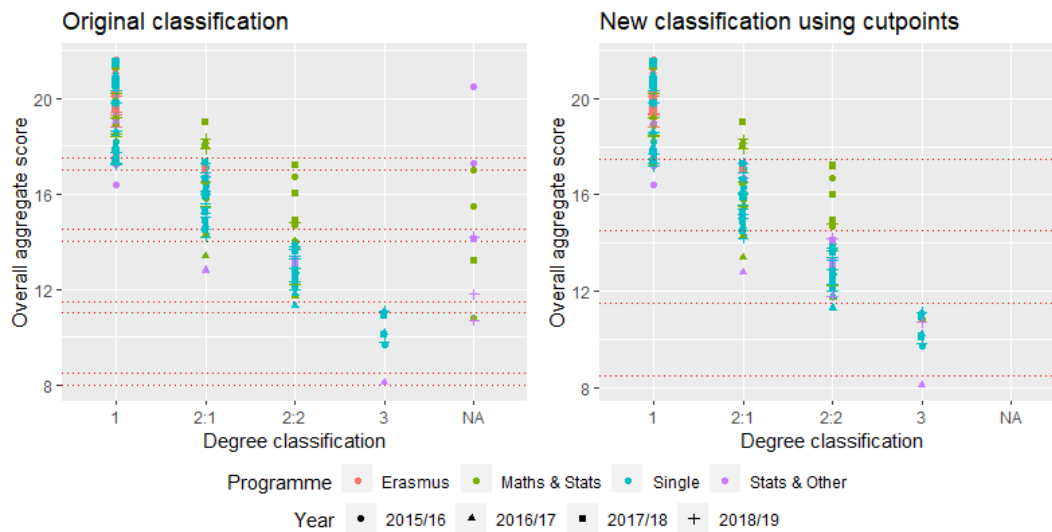


FIGURE 3.1: Relationship between overall aggregate score and original degree classification (left), new degree classification (right)

These cutpoints are slightly different from the criteria in Figure A.2 and A.3 because we do not have enough information to give the results precisely. The relationship after filling degree classification are shown in Figure 3.1 (right). Red dashed lines in the figure are the cutpoints of each classification.

Lastly, we create two new data frames transforming from course results to analyse. One is to transform secondary band scores to a 22-point numerical score A.1, where we transform "MV" (good cause from medical or personal circumstances) to "NA" (not available). We use this transformation in question 1, 2, and 3. Another is to transform the secondary band score to enrolment state (True for enrol and False for not enrol). This is for question 5.

3.2 Are there differences in the distribution of course results?

We have 28 courses in our data; all are in Section 1.4. We will look at the distribution of results in this question.

3.2.1 Exploratory data analysis

Some courses contain only a few numbers of data, for example, S003 and S010. A complete table of number is shown in Table 3.1. The best way to compare the distribution of every course is to use boxplots. Figure 3.2 (top) shows such distributions.

S010	S003	S013	S012	S017	S024	S011	S029	S027	S028
1	1	2	3	3	4	5	5	9	10
S018	S009	SProj.20Cr	S005	S023	S007	S019	S006	S016	
12	24	25	39	40	42	47	68	76	
S008	S022	SProj.30Cr	S002	S014	S026	S001	S025	S021	
110	110	111	117	117	118	137	146	150	

TABLE 3.1: Number of students enrolling in each course

Some courses are also selected by all student in some programmes, or some years, while some courses are not compulsory. Figure B.1 to B.8 and Table C.1 to C.2 shows the distribution of students in every programme and year. We can see that some courses have never been selected.

3.2.2 Analyses and model checks

We perform F-tests and Student's t-tests to test the variance and mean equality between each pair of courses. Firstly, we will perform F-test by

1. Only courses with more than three observations are selected
2. Assuming that each course result follows a normal distribution
3. Assuming that course results from the same student are independent

There are 23 out of 28 courses having observations number greater than three in Table 3.1. We have $\binom{23}{2} = 253$ pairs of course combinations. The results from programming indicate that 121 pairs fail to accept the null hypothesis at a 95% confidence level. An example of a high variance difference is between S002 and S008.

After cutting half of the pairs having different variances, we assume that each course result has the same variance as we do not have enough evidence to reject. We use Student's t-tests on the other pairs (132 pairs). Only 41 pairs fail to accept the null hypothesis at a 95% confidence level. An example of a high mean difference is between S014 and S008.

3.2.3 Interpretations of analyses

After testing the variance, we find that 47.83% of the pairs of courses have different variances. On the other hand, 31.06% of those with no differences have different means.

In the sense of median, we can see differences when we sort the box by their value in Figure 3.2 (bottom).

We also notice that outliers appear only on the left. We expect to see this phenomenon due to the natural aspect of studying; getting a low score can be more effortless than a high. Most distributions are not right-skewed, except for S027, S012, S028, and S029. Many courses have students who get the possible highest score - A1 or 22.

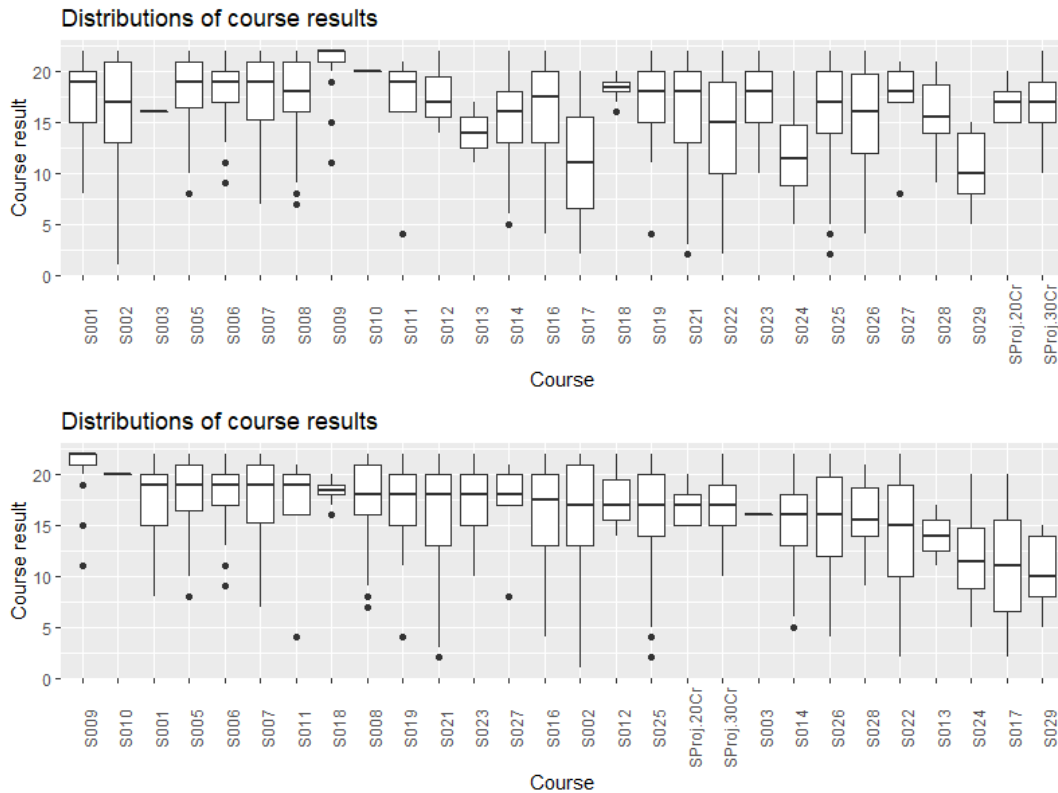


FIGURE 3.2: Distribution of course results, alphabetically (top)
Distribution of course results, sorted by median (bottom)

3.2.4 Conclusions for specific questions

In conclusion, there are differences in the distribution in many ways - mean, median and shape. Most of them are normally distributed or left-skewed, and the outliers always occur on the left.

3.3 Are there differences in the course and degree results between the different programmes of study?

There are four different programmes of study in our data as described in Section 1.4. We investigate the course and degree results between different programmes of study in this section.

3.3.1 Exploratory data analysis

As we know from the last section, some programmes of study do not choose some courses. However, some courses are compulsory in some programmes. A few courses are selected by students in all programmes, for example, course S001. A full table is Table C.1.

Programme	Degree Classification			
	1	2:1	2:2	3
Erasmus	19 0.95	1 0.05	0 0	0 0
Maths & Stats	16 0.33	15 0.31	16 0.33	1 0.02
Single	32 0.35	27 0.30	25 0.27	7 0.08
Stats & Other	9 0.47	3 0.16	5 0.26	2 0.11

TABLE 3.2: Contingency table of programme and degree classification

The contingency table of degree results by programme can be shown in Table 3.2. In Figure 3.3 (left), the majority degree classification of Erasmus student is first class. The others share the proportion equally. The missing values are the values we have explained. To compare the proportion, we see from Figure 3.3 (right). However, we do not see a significant difference between Maths & Stats and Single students, but we do for the other pairs. We conduct statistical tests.

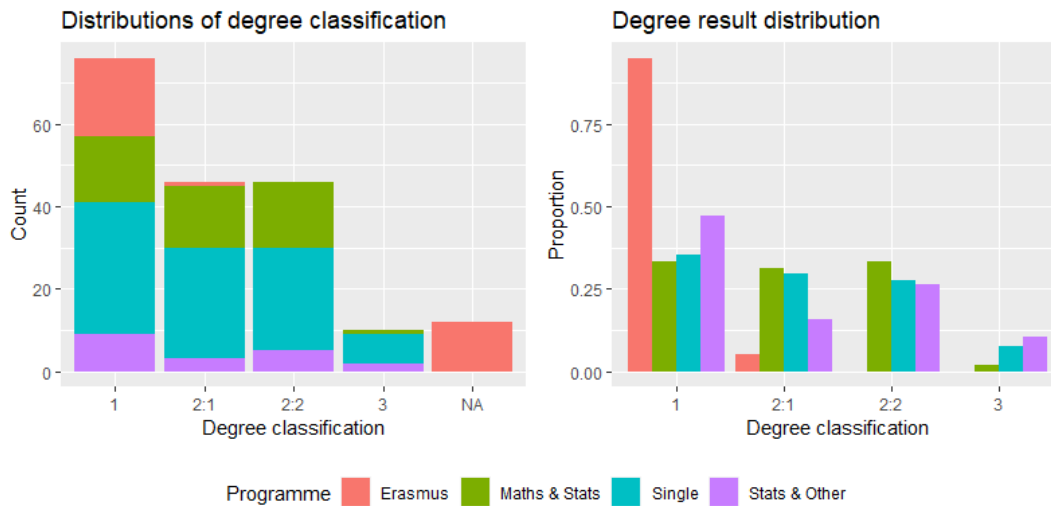


FIGURE 3.3: Distribution of degree classification by programme of study

3.3.2 Analyses and model checks

For course results, we perform F-test to test the equality of the variance of each course by

1. Considering programmes with more than three students enrolling in that selected course
2. Assuming that each course result follows a normal distribution

There are only 54 pairs of programmes and courses that satisfy the first criteria. Only 13 of them are significantly different in variance at 95% confidence level. The pair of Erasmus and single programme students in S001 is an example.

The rest 41 pairs have no difference in variance. We use Student's t-tests to test the equality of the mean between programmes under the assumption of normality and equality of variance. Only nine pairs have significant significant mean differences at a 95% confidence level. An example is the pair of Erasmus and Math & Stats programme students in S014.

We use Pearson's Chi-squared test for degree results under the null hypothesis that the degree classification and programme are independent. Since we see extreme proportion in Figure 3.3 (right), we conduct test in three cases

1. All programmes against all degree classifications
We have $\chi^2 = 30.2790$ with a degree of freedom 9. The p-value is 0.0004.
2. All programmes except Erasmus against all degree classifications
We have $\chi^2 = 4.6336$ with a degree of freedom 6. The p-value is 0.5916.
3. All programmes against all degree classifications except first class degree
We have $\chi^2 = 4.8439$ with a degree of freedom 6. The p-value is 0.5640.

3.3.3 Interpretations of analyses

Due to the small sample size, 114 out of $\binom{4}{2} \times 28 = 168$ combinations (67.86%) fail to perform F-test under our first assumption. 24.07% of the rest have a significant variance difference at a 95% confidence level. For those who do not have, 21.95% of them have significant mean differences at the same confidence level. We conclude that there are some differences in course results between programmes.

On the other hand, we see that degree classifications and degree results are significantly dependent. From Figure 3.3 (right), we see the apparent evidence from Erasmus student in the first class degree classification. When we ignore Erasmus students or first degree classification, we cannot deny their independence.

3.3.4 Conclusions for specific questions

In conclusion, there are some differences in the course results in mean and variance. There are significant differences in degree results if we consider Erasmus students and first class degree classification together.

3.4 Are there any differences in the course and degree results between years of graduation?

Our data contains four years of graduation as described in Section 1.4. We investigate the course and degree results between different years of graduation in this section.

3.4.1 Exploratory data analysis

Similar to the last question, some courses are not chosen in some years of graduation. A few courses are selected by students every year; for example, course S001, as shown in Figure B.10. A full table can be seen in Table C.2.

The distribution of degree results by year is in Table 3.3. In Figure 3.4 (left), the first class degree of 2015/16 and 2016/17 seems to have a bigger number than other degree classifications. The others share the proportion equally. The missing values are Erasmus students who do not have degree classifications. To compare the proportion, we see from Figure 3.4 (right). We do not see a huge difference in proportion, so we conduct statistical tests.

Year	Degree Classification			
	1	2:1	2:2	3
2015/16	25 0.51	11 0.22	11 0.22	2 0.04
2016/17	20 0.53	8 0.21	7 0.18	3 0.08
2017/18	15 0.35	11 0.26	15 0.35	2 0.05
2018/19	16 0.33	16 0.33	13 0.27	3 0.06

TABLE 3.3: Contingency table of year and degree classification

3.4.2 Analyses and model checks

For course results, we perform F-test to test the equality of the variance of each course by

1. Only year with more than three students enrolling in that selected course are considered
2. Assuming that each course result follows a normal distribution

There are 89 pairs of programmes and courses that satisfy the first criteria. Only 13 are significantly different in variance at a 95% confidence level. The pair of the graduation year 2017/18 and 2018/19 in S005 is an example of those.

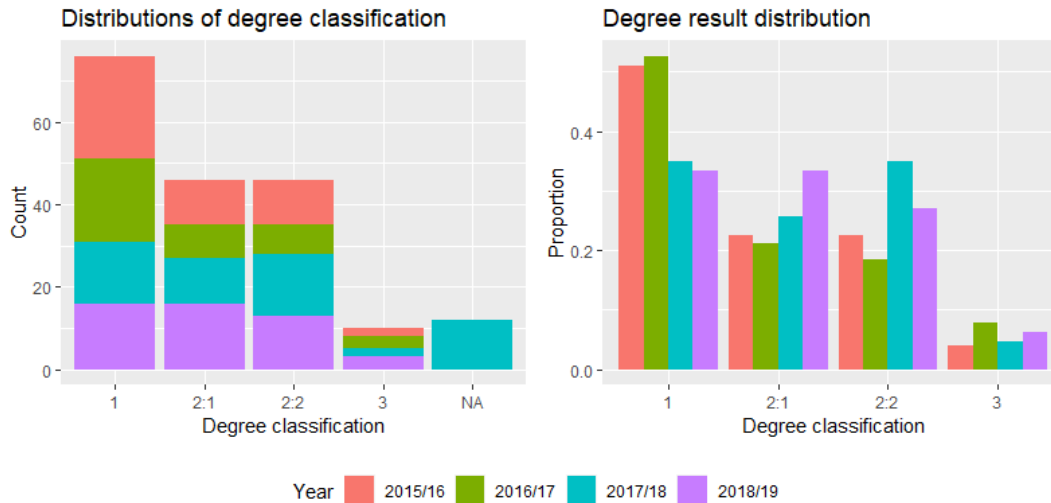


FIGURE 3.4: Distribution of degree classification by graduation year

For the rest of them, 76 pairs having no difference in variance, we use Student's t-tests to test the equality of the mean between programmes under the assumption of normality and equality of variance. Only 11 pairs have significant mean differences at a 95% confidence level. The pair of the graduation year 2017/18 and 2018/19 in S001 is an example of those.

We use Pearson's Chi-squared test for degree results under the null hypothesis that the degree classification and graduation year are independent. Since we see different proportions in Figure 3.4 (right), we conduct test in four cases

1. All graduation years against all degree classifications
We have $\chi^2 = 7.9522$ with a degree of freedom 9. The p-value is 0.5390.
2. Graduation year 2017/18 and 2018/19 against all degree classifications
We have $\chi^2 = 1.0294$ with a degree of freedom 3. The p-value is 0.7941.
3. All graduation years against all degree classifications except first class degree
We have $\chi^2 = 2.2552$ with a degree of freedom 6. The p-value is 0.8948.
4. All graduation years against all degree classifications except third class degree
We have $\chi^2 = 7.2745$ with a degree of freedom 6. The p-value is 0.2962.

3.4.3 Interpretations of analyses

Due to the small sample size, 79 out of $\binom{4}{2} \times 28 = 168$ combinations (47.02%) fail to perform F-test under our first assumption. 14.61% of the rest have significant differences in variance at a 95% confidence level. For those who do not have, 14.47% of them have significant mean differences at the same confidence level. We conclude that there are some differences in course results

between programmes. On the other hand, we do not have enough evidence to reject their independence in all cases.

3.4.4 Conclusions for specific questions

In conclusion, there are some differences in the course results in mean and variance. There are no significant differences in degree results.

3.5 What is the relationship between the aggregate result from third year statistics courses and the overall aggregate score and class of degree awarded?

In this question, the missing values from the third year are omitted instead of replacing the value with the fourth year or aggregate results.

3.5.1 Exploratory data analysis

The relationship between third year and overall aggregate score and the relationship between third year and class of degree awarded are shown on the left and right of Figure 3.5, respectively.

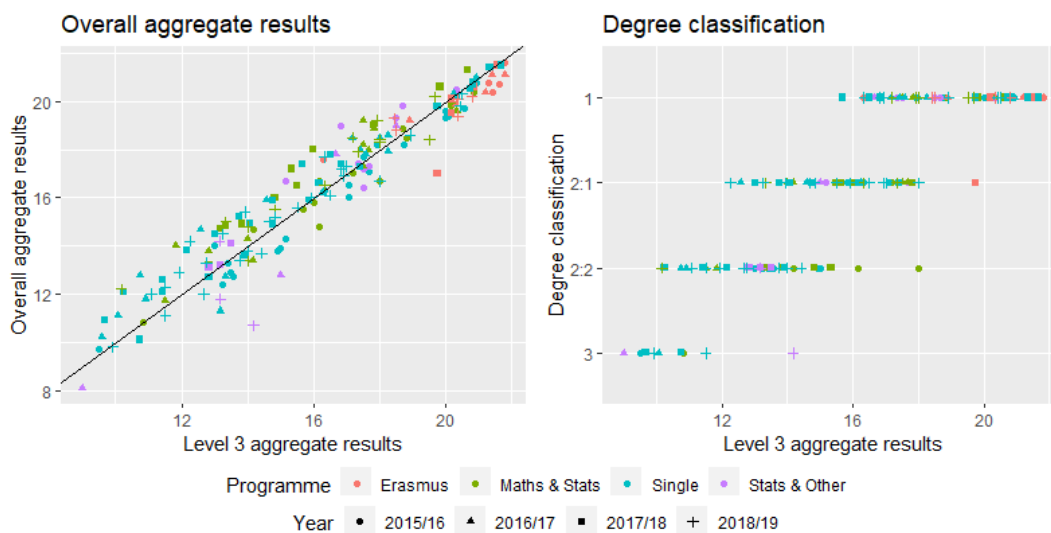


FIGURE 3.5: Relationship between level 3 results and overall aggregate results (left) and degree classification (right)

3.5.2 Analyses and model checks

It is interesting to fit this relationship to a linear model,

$$\text{Overall_aggregate} \sim \text{Level3_aggregate} + \text{Programme} + \text{Year}$$

We see the model assumptions hold in Figure D.1. The residuals are scattered around zero and seem to follow a normal distribution except for both ends. The modulus of standardised residuals decreases as the fitted score increases. These are acceptable and do not violate the assumptions. The equation is

$$\begin{aligned} \text{Overall_aggregate}_i = & \beta_0 + \beta_1 \text{Level3_aggregate}_i + \beta_2 \mathbb{I}_{\text{Programme:Maths\&Stats},i} + \\ & \beta_3 \mathbb{I}_{\text{Programme:Single},i} + \beta_4 \mathbb{I}_{\text{Programme:Stats\&Other},i} + \\ & \beta_5 \mathbb{I}_{\text{Year:2016/17},i} + \beta_6 \mathbb{I}_{\text{Year:2017/18},i} + \beta_7 \mathbb{I}_{\text{Year:2018/19},i} + \epsilon_i \end{aligned}$$

where $\epsilon_i \sim N(0, \sigma^2)$. The coefficients of this model are given in Table 3.4.

Coefficients	Estimate	Std. Error	P-value
β_0	0.7700	0.5324	0.1499
β_1	0.9276	0.0226	< 0.0001
β_2	0.5844	0.2469	0.0190
β_3	0.2410	0.2385	0.3137
β_4	0.1340	0.2990	0.6547
β_5	0.3697	0.1889	0.0520
β_6	0.6522	0.1839	0.0005
β_7	0.2607	0.1803	0.1501

TABLE 3.4: Coefficients for
Overall_aggregate $\sim$ Level3_aggregate + Programme + Year

For the degree classification, we use `polr` function from library `MASS` to fit the proportional odd logistic model. For the i^{th} observation, the model has three equations;

$$\begin{aligned} \log \left(\frac{p_{1,i}}{p_{2:1,i} + p_{2:2,i} + p_{3,i}} \right) = & \beta_{0,1} - \beta_1 \text{Level3_aggregate}_i - \beta_2 \mathbb{I}_{\text{Programme:Maths\&Stats},i} - \\ & \beta_3 \mathbb{I}_{\text{Programme:Single},i} - \beta_4 \mathbb{I}_{\text{Programme:Stats\&Other},i} - \\ & \beta_5 \mathbb{I}_{\text{Year:2016/17},i} - \beta_6 \mathbb{I}_{\text{Year:2017/18},i} - \beta_7 \mathbb{I}_{\text{Year:2018/19},i} \\ \log \left(\frac{p_{1,i} + p_{2:1,i}}{p_{2:2,i} + p_{3,i}} \right) = & \beta_{0,2} - \beta_1 \text{Level3_aggregate}_i - \beta_2 \mathbb{I}_{\text{Programme:Maths\&Stats},i} - \\ & \beta_3 \mathbb{I}_{\text{Programme:Single},i} - \beta_4 \mathbb{I}_{\text{Programme:Stats\&Other},i} - \\ & \beta_5 \mathbb{I}_{\text{Year:2016/17},i} - \beta_6 \mathbb{I}_{\text{Year:2017/18},i} - \beta_7 \mathbb{I}_{\text{Year:2018/19},i} \\ \log \left(\frac{p_{1,i} + p_{2:1,i} + p_{2:2,i}}{p_{3,i}} \right) = & \beta_{0,2} - \beta_1 \text{Level3_aggregate}_i - \beta_2 \mathbb{I}_{\text{Programme:Maths\&Stats},i} - \\ & \beta_3 \mathbb{I}_{\text{Programme:Single},i} - \beta_4 \mathbb{I}_{\text{Programme:Stats\&Other},i} - \\ & \beta_5 \mathbb{I}_{\text{Year:2016/17},i} - \beta_6 \mathbb{I}_{\text{Year:2017/18},i} - \beta_7 \mathbb{I}_{\text{Year:2018/19},i} \end{aligned}$$

where $p_{1,i} + p_{2:1,i} + p_{2:2,i} + p_{3,i} = 1$, each variable represents the probability of having the subscripted degree. The coefficients of this model are given in Table 3.5. When we predict the class on the highest probability, the accuracy

of this model is 136 out of 178, equivalently, 76.40%. The Brier Score of this prediction is 0.0736.

Coefficients	Estimate	Std. Error	P-value
$\beta_{0,1}$	-24.6435	3.4458	< 0.0001
$\beta_{0,2}$	-20.6218	3.1098	< 0.0001
$\beta_{0,3}$	-15.6016	2.7845	< 0.0001
β_1	-1.4261	0.1634	< 0.0001
β_2	1.4222	1.7464	0.4155
β_3	0.2799	1.7377	0.8720
β_4	0.1765	1.7966	0.9217
β_5	-1.8768	0.6209	0.0025
β_6	-0.8581	0.5805	0.1394
β_7	-0.9871	0.5415	0.0683

TABLE 3.5: Coefficients for the proportional odd logistic model
Classification $\sim$ Level3_aggregate + Programme + Year

3.5.3 Interpretations of analyses

For the relationship of overall aggregate score, the significant variables are level 3 aggregate score, Maths & Stats programme, and graduation year 2017/18 on the baseline categories of Erasmus programme and graduation year 2015/16. Every one score of level 3 aggregate will lead to 0.93 increments on the overall aggregate score. On the other hand, in the relationship of the class of degree awarded, every one increment of level 3 aggregate score will multiply the odd by 4.1624 ($e^{1.4261}$). We see some differences between intercept terms as we expect, but the confidence interval at a 95% level are overlapping. Another significant variable is the graduation year 2016/17, which multiplies the odd by 6.5326 ($e^{1.8768}$). In short, the negative coefficients of non-intercept terms in the table make the probability of getting a better degree classification higher, while the positive ones make it lower.

3.5.4 Conclusions for specific questions

There are significant linear relationships in both of them. The only significant variable in both relationships is the graduation year 2017/18. As a classification tool for the class of degree awarded, the model can accurately predict up to approximately 75%.

3.6 What is the relationship between the choice of final year statistics courses and the overall aggregate score and class of degree awarded? If there is a relationship, which courses should be chosen in the final year of study to maximize the chances of a first class degree?

Each course's score relates to the aggregate score by a formula. We investigate the course combination instead.

3.6.1 Exploratory data analysis

In Table 3.1, we see the number of students enrolling in each course.

3.6.2 Analyses and model checks

We can use a linear model to answer the choice combination problem. There are two possible approaches we are considering:

1. A linear model with no course interactions

$$\text{Overall_aggregate} \sim \text{Course} + \dots + \text{Course}^1$$
2. A linear model with course interactions

$$\text{Overall_aggregate} \sim (\text{Course} + \dots + \text{Course}) \times (\text{Course} + \dots + \text{Course})$$

Their model assumption is shown in Figure D.2 and D.3, respectively. The residual means are zero, and the residuals follow normal distributions. However, when the fitted score is around 16, the modulus of standardised residuals is at its peak and higher than the rest fitted values.

Table 3.6 shows some interesting coefficients in the first model. For a full table, see Table E.1. We observe that only one variable is significant at a 95% confident level. For the second model, there is only one more significant variable to the first model. Similarly, Table 3.7 shows some interesting variables. For a full table, see Table E.2 and E.3.

Instead, suppose we change the overall aggregate result to level 4 aggregate result in the model;

3. A linear model with no course interactions

$$\text{Level4_aggregate} \sim \text{Course} + \dots + \text{Course}$$
4. A linear model allowing course interactions

$$\text{Level4_aggregate} \sim (\text{Course} + \dots + \text{Course}) \times (\text{Course} + \dots + \text{Course})$$

¹Course + ... + Course in the model refers to S001 + S002 + ... + SProj.30Cr, all 28 listed courses in Section 1.4. In this notation, they are binary explanatory variables and *not* continuous in the form of a 22 band score. For each student, each course takes the value TRUE (enrol) or FALSE (not enrol). We sometimes denote by $\sum \text{Course}$ for convenience.

Coefficients	Estimate	Std. Error	P-value
Intercept	0.5036	13.4454	0.9702
S009:TRUE	5.7319	2.3224	0.0147

TABLE 3.6: Some coefficients for
Overall_aggregate $\sim \sum$ Course

Coefficients	Estimate	Std. Error	P-value
Intercept	93.9017	139.4064	0.5020
S007:TRUE and S001:TRUE	35.2242	14.8111	0.0191
S002:TRUE and S001:TRUE	-40.1311	19.7425	0.0444

TABLE 3.7: Some coefficients for
Overall_aggregate $\sim \sum$ Course $\times \sum$ Course

In this case, we obtain some more significant coefficients, they are in Table 3.8 (See all in Table E.4) and Table 3.9 (See all in Table E.5 and E.6). The model assumption of each model is shown in Figure D.4 and D.5. We do not see much difference compared to the first two models.

Another question is, how can we maximise the chances of a first class degree. If students expect to have that degree classification, it is wise to look for combinations of first class students. We find no significant variables when we observe the first two linear models on first class students. Instead, we find some if we change the model from overall to level 4 aggregate score. There are interesting results from the model with non-interaction in Table 3.10. For the interactions model, no variables are significant. All coefficients of all models are in Table E.7 to E.10.

For the relationship between the choice of final year statistics courses and the class of degree awarded, we use `polr` function from library `MASS` to fit the proportional odd logistic model and find it does not converge. It is worth mentioning a `multinom` function from library `nnet` that can be used fit a multinomial model. This model predicts correctly 101 out of 178 (56.74%). The brier score of this prediction is 0.1290.

3.6.3 Interpretations of analyses

In the first two models, we see high standard errors in the intercept term, showing that the models are not entirely stable. However, we can reduce those standard errors when we use the last two models instead, and the intercept terms become significant. Things these models share in common are S009 itself, and S001 combined with S007 significantly lead to a higher aggregate score. If we consider separately, SProj.20Cr significantly leads to a higher level 4 aggregate score, while S011 and S017 lead to a lower level 4

Coefficients	Estimate	Std. Error	P-value
Intercept	13.3111	1.7591	< 0.0001
S009:TRUE	3.0226	0.9354	0.0015
SProj.20Cr:TRUE	2.7979	1.0319	0.0074
S017:TRUE	-6.2272	2.1592	0.0045
S011:TRUE	-3.3192	1.5137	0.0298

TABLE 3.8: Significant coefficients (95%) for
Level4_aggregate $\sim \sum \text{Course}$

Coefficients	Estimate	Std. Error	P-value
Intercept	8.4778	3.9154	0.0325
S001:TRUE and S007:TRUE	34.0556	15.1564	0.0267

TABLE 3.9: Significant coefficients (95%) for
Level4_aggregate $\sim \sum \text{Course} \times \sum \text{Course}$

Coefficients	Estimate	Std. Error	P-value
S023TRUE	3.9457	1.8899	0.0415
S009TRUE	3.7507	1.8708	0.0500

TABLE 3.10: Significant coefficients (95%) for
Level4_aggregate $\sim \sum \text{Course}$
(first class students)

aggregate score, and S001 combined with S002 lead to a lower overall aggregate score. The significant courses that enhance the chance of getting first class degree are S009 and S023.

The reason that there are few significant courses is that the result of studying is individual. Students choosing the same course can have highly different results. We see from the multinomial model as a classification tool that only 56.74% predict the result correctly from choices of courses each student choose.

3.6.4 Conclusions for specific questions

In conclusion, there is some relationship. In order to maximize the chances of a first class degree, we suggest taking S009, SProj.20Cr, S001, S007, and S023. If students consider enrolling in additional courses, they should be careful because some combinations can lead to a worse result. However, the analysis shows that the study result is totally individual.

Chapter 4

Conclusions and Discussion

4.1 Conclusion

From what we have analysed in the previous chapter, we conclude that there is a relationship between choices in final year and class of honours degree.

First, when we consider the overall distribution of each course result, we can see differences in mean, median and variance. Most follow a normal distribution or are left-skewed, and the outliers always occur on the left. After that, we consider the course results between each programme of study and each graduation year. We observe that there are some differences in mean and variance, however.

On the other hand, if we consider the degree results, there are significant differences when Erasmus students and first class degree classification are considered together. When we ignore one of them, we do not see significant differences. We also observe no significant differences in degree results between graduation years.

Significant linear relationships exist between the aggregate result from third year statistics courses and between the overall aggregate score and class of degree awarded. The only significant variable in both relationships is the graduation year 2017/18. We can correctly predict three out of four of the degree awarded classification using only the information from third year statistics.

Lastly, there is some relationship between the choice of final year statistics courses and the overall aggregate score and class of degree awarded. The analysis shows that the study result is individual and likely not to depend on choices of courses. To maximise the chances of a first class degree, we suggest taking S009, SProj.20Cr, S001, S007, and S023.

4.2 Future Work

If we have the conditions of course in each year to enrol and comments of each student result that behave differently, it would be beneficial to the analysis. Also, the data size is small: we have only less than 200 observations available. If we could have the results of each level 3 course, we should be able to analyse the combination problem thoroughly.

Appendix A

Guide to the Code of Assessment

The University of Glasgow, *Guide to the Code of Assessment - Chapter 2 2022*, have provided information on the grading system. Some tables from the guide are shown in this appendix.

SCHEDULE A				
Primary Grade	Gloss	Secondary Band*	Grade Point	Primary Verbal Descriptors for Attainment of Intended Learning Outcomes
A	Excellent	A1	22	Exemplary range and depth of attainment of intended learning outcomes, secured by discriminating command of a comprehensive range of relevant materials and analyses, and by deployment of considered judgement relating to key issues, concepts and procedures
		A2	21	
		A3	20	
		A4	19	
		A5	18	
B	Very Good	B1	17	Conclusive attainment of virtually all intended learning outcomes, clearly grounded on a close familiarity with a wide range of supporting evidence, constructively utilised to reveal appreciable depth of understanding
		B2	16	
		B3	15	
C	Good	C1	14	Clear attainment of most of the intended learning outcomes, some more securely grasped than others, resting on a circumscribed range of evidence and displaying a variable depth of understanding
		C2	13	
		C3	12	
D	Satisfactory†	D1	11	Acceptable attainment of intended learning outcomes, displaying a qualified familiarity with a minimally sufficient range of relevant materials, and a grasp of the analytical issues and concepts which is generally reasonable, albeit insecure
		D2	10	
		D3	9	
E	Weak	E1	8	Attainment deficient in respect of specific intended learning outcomes, with mixed evidence as to the depth of knowledge and weak deployment of arguments or deficient manipulations
		E2	7	
		E3	6	
F	Poor	F1	5	Attainment of intended learning outcomes appreciably deficient in critical respects, lacking secure basis in relevant factual and analytical dimensions
		F2	4	
		F3	3	
G	Very Poor	G1	2	Attainment of intended learning outcomes markedly deficient in respect of nearly all intended learning outcomes, with irrelevant use of materials and incomplete and flawed explanation
		G2	1	
H			0	No convincing evidence of attainment of intended learning outcomes, such treatment of the subject as is in evidence being directionless and fragmentary
CR	CREDIT REFUSED			Failure to comply, in the absence of good cause, with the published requirements of the course or programme; and/or a serious breach of regulations

FIGURE A.1: Relationship between secondary bands and grade points

17.5 to 22.0	First class honours
14.5 to 17.0	Upper second class honours
11.5 to 14.0	Lower second class honours
8.5 to 11.0	Third class honours
0.0 to 8.0	Fail

FIGURE A.2: Relationship between class degree awards and grade points

17.1 to 17.4	First class honours: where at least 50% of the weighted course grade profile comprises A grades
	Upper second class honours: where less than 50% of the weighted course grade profile comprises A grades
14.1 to 14.4	Upper second class honours: where at least 50% of the weighted course grade profile comprises B grades
	Lower second class honours: where less than 50% of the weighted course grade profile comprises grades of B or above
11.1 to 11.4	Lower second class honours: where at least 50% of the weighted course grade profile comprises grades of C or above
	Third class honours: where less than 50% of the weighted course grade profile comprises grades of C or above
8.1 to 8.4	Third class honours: where at least 50% of the weighted course grade profile comprises grades of D or above
	Fail: where less than 50% of the weighted course grade profile comprises grades of D or above.

FIGURE A.3: Relationship between class degree awards and grade points (cont.)

Appendix B

Course Result Distributions

The course results are integers ranging from 0 to 22 after converting the secondary band to grade point using Figure [A.1](#). There are 28 courses, four programmes of study and four years of graduation in total.

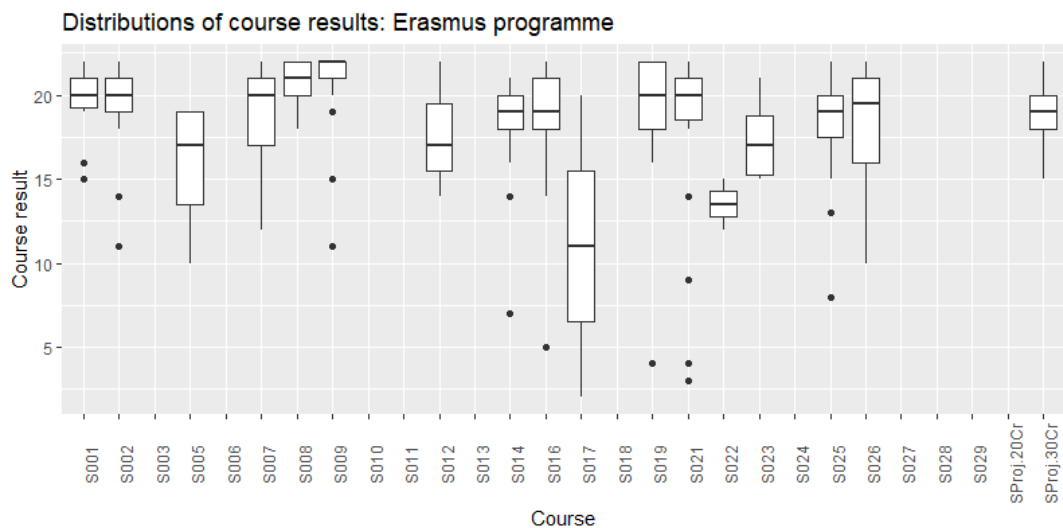


FIGURE B.1: Score distribution of students in Erasmus programme

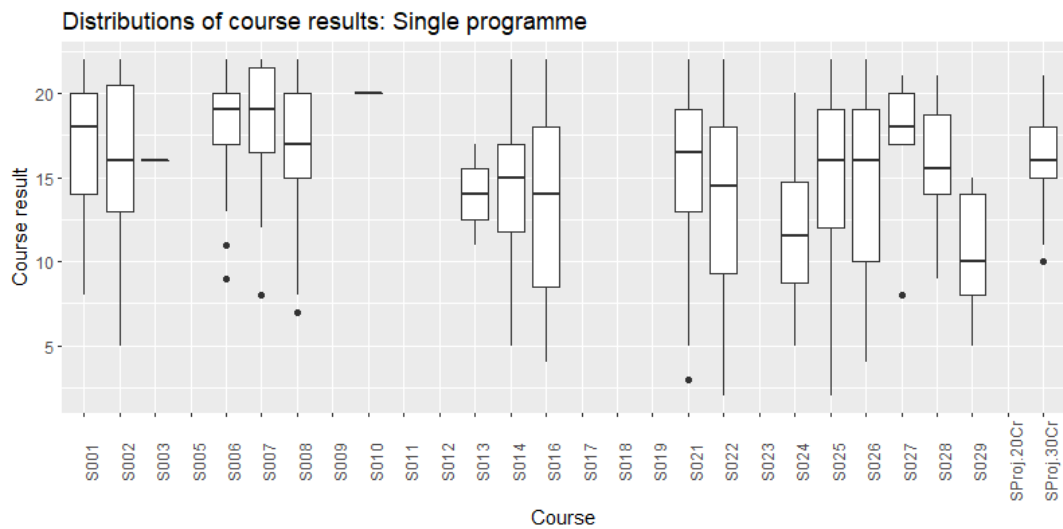


FIGURE B.2: Score distribution of students in Single programme

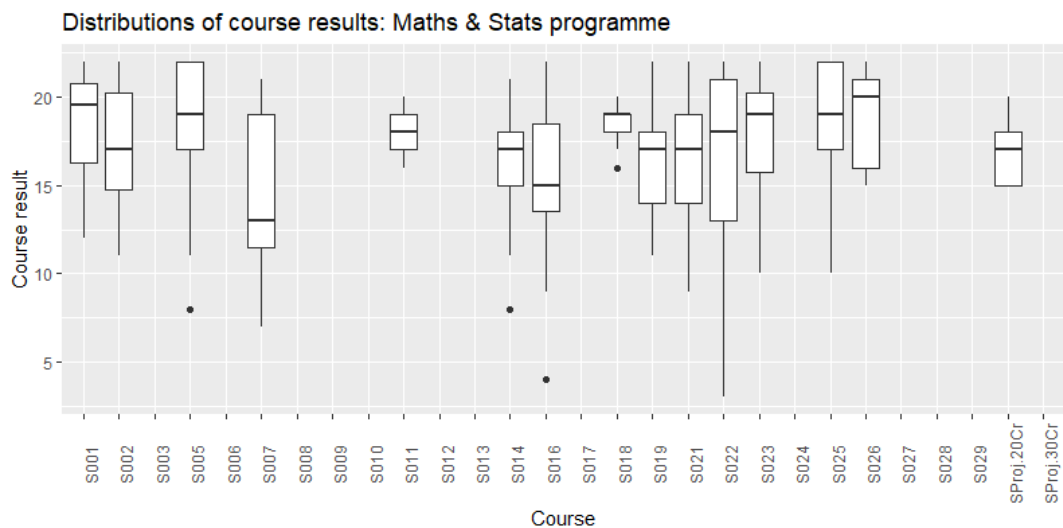


FIGURE B.3: Score distribution of students in Maths & Stats programme

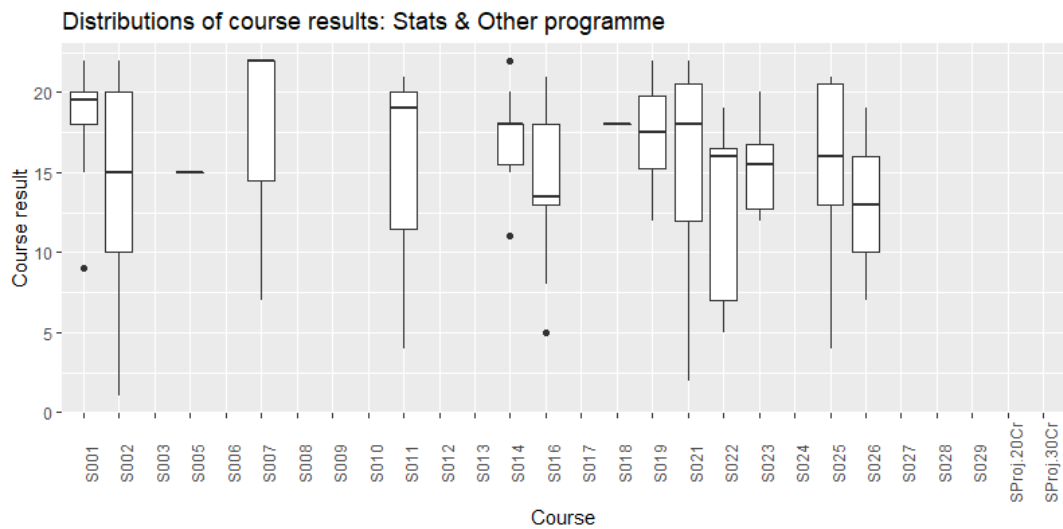


FIGURE B.4: Score distribution of students in Stats & Other programme

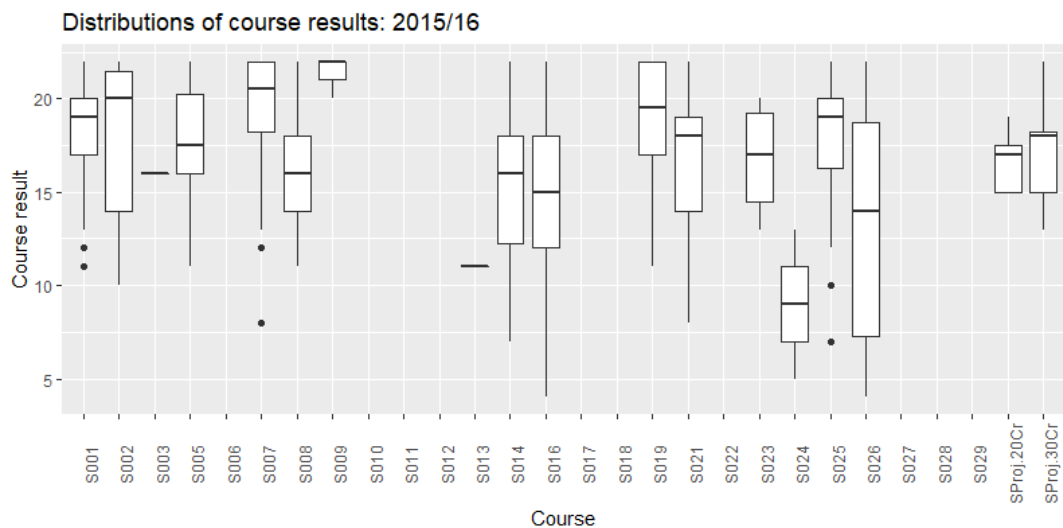


FIGURE B.5: Score distribution of students in 2015/16 graduation year

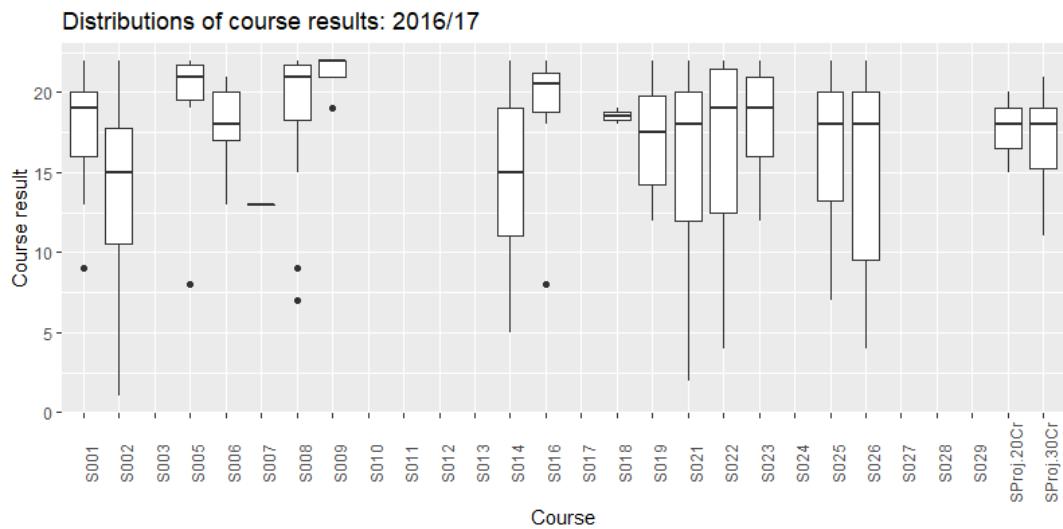


FIGURE B.6: Score distribution of students in 2016/17 graduation year

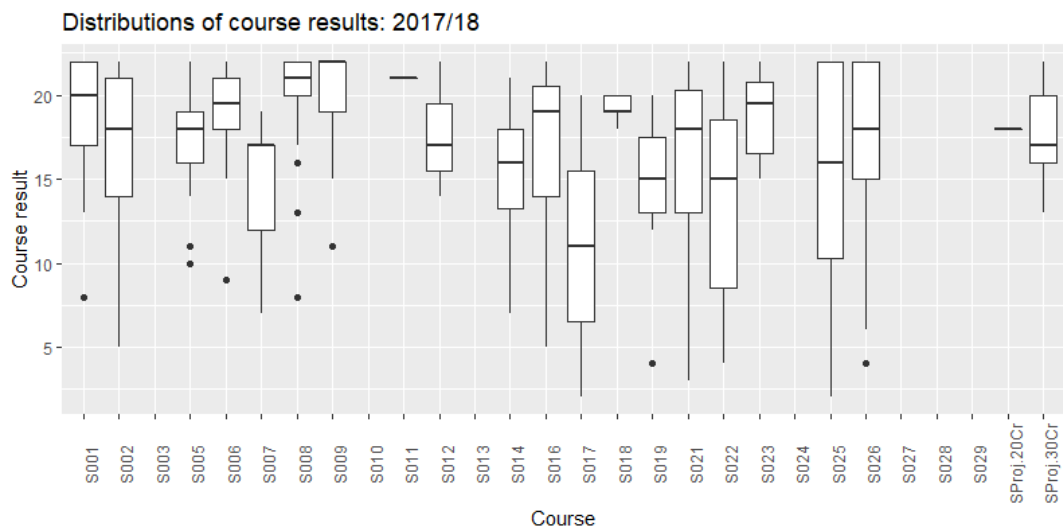


FIGURE B.7: Score distribution of students in 2017/18 graduation year

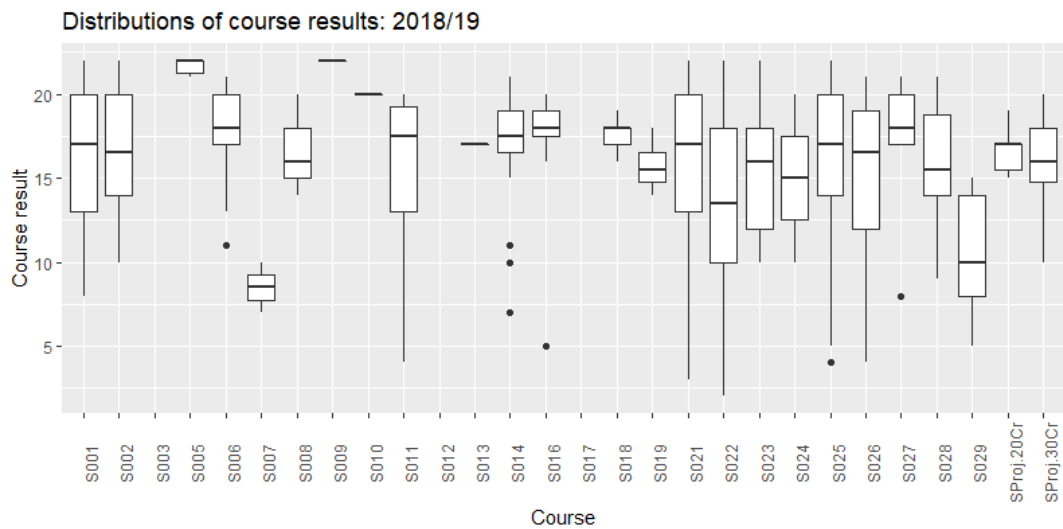


FIGURE B.8: Score distribution of students in 2018/19 graduation year

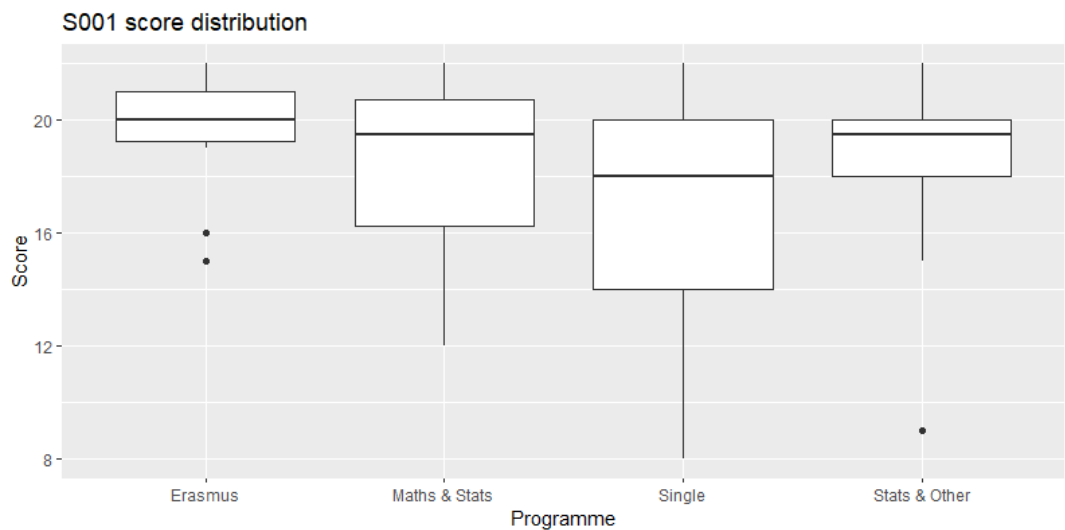


FIGURE B.9: Score distribution of S001 in each programme of study

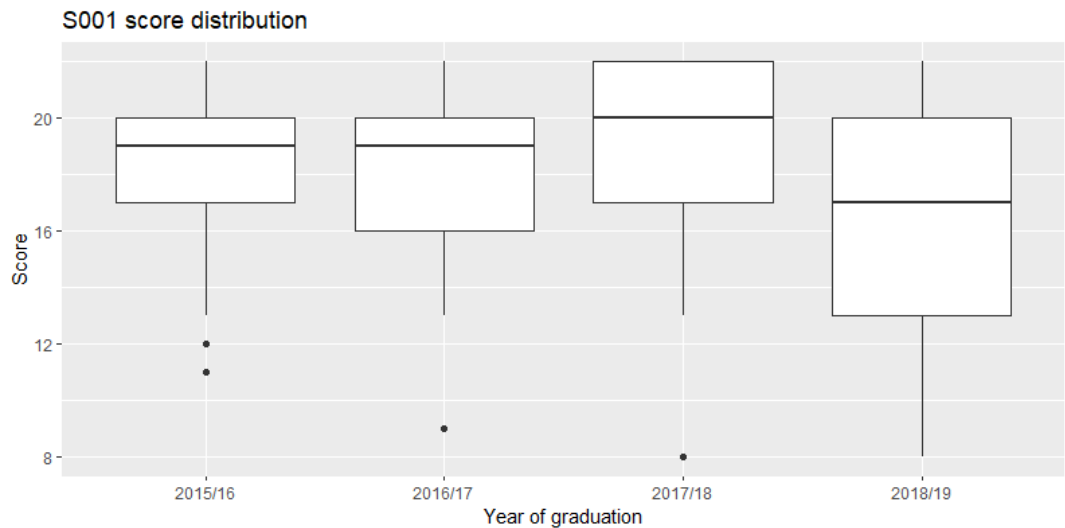


FIGURE B.10: Score distribution of S001 in each year of study

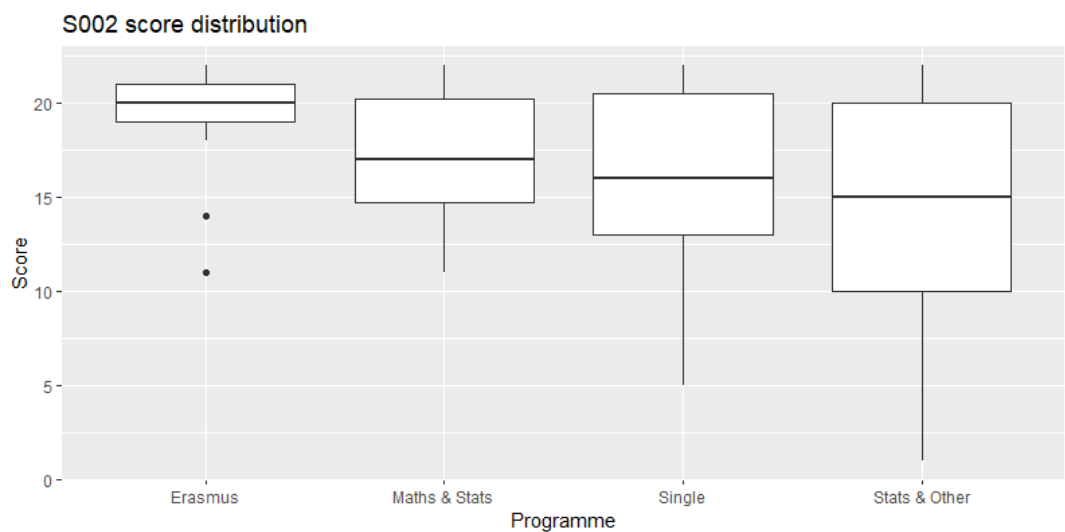


FIGURE B.11: Score distribution of S002 in each programme of study

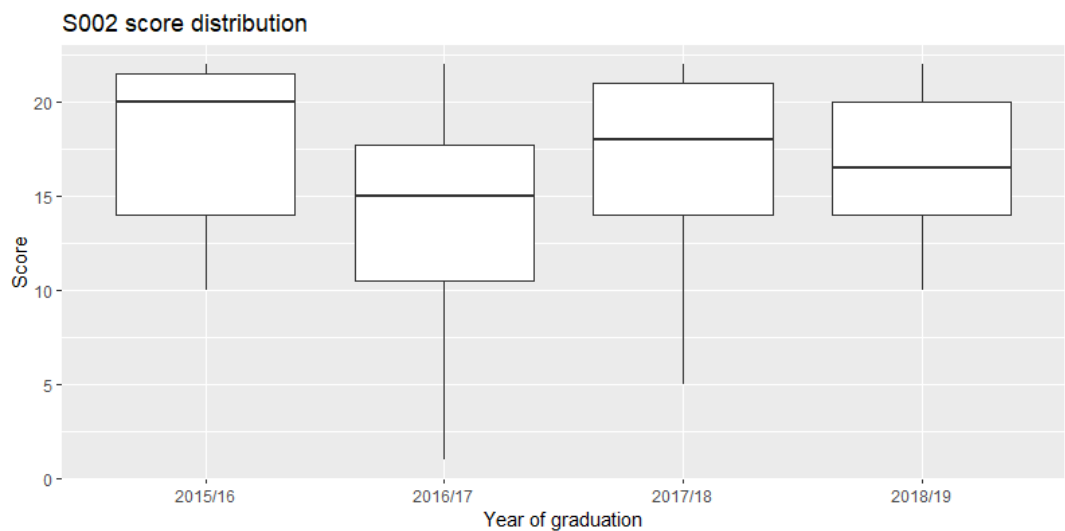


FIGURE B.12: Score distribution of S002 in each year of study

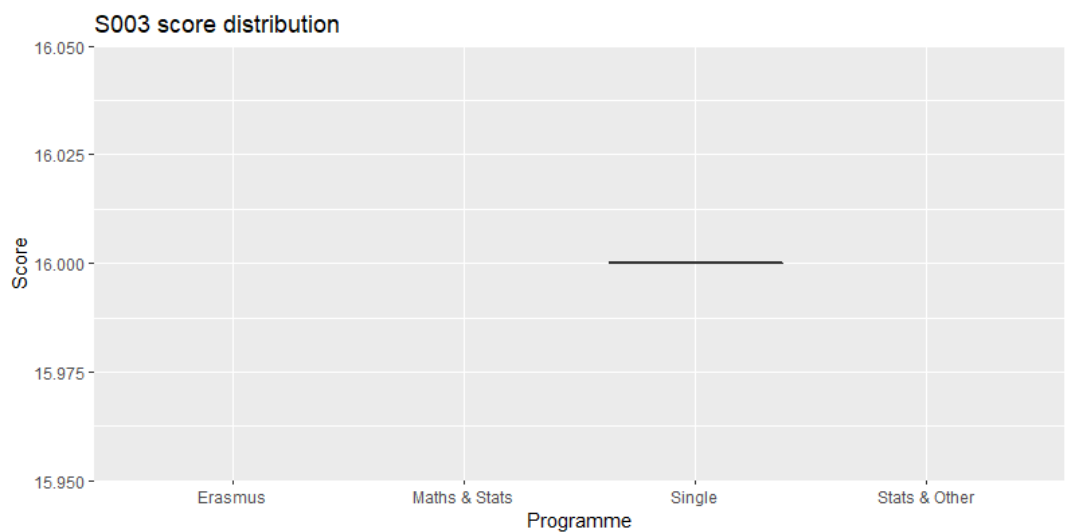


FIGURE B.13: Score distribution of S003 in each programme of study

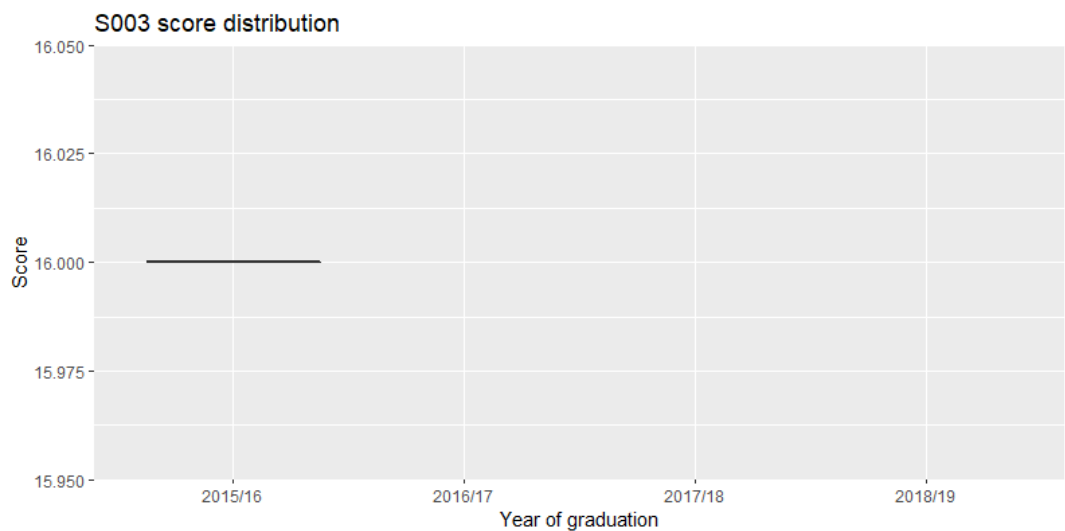


FIGURE B.14: Score distribution of S003 in each year of study

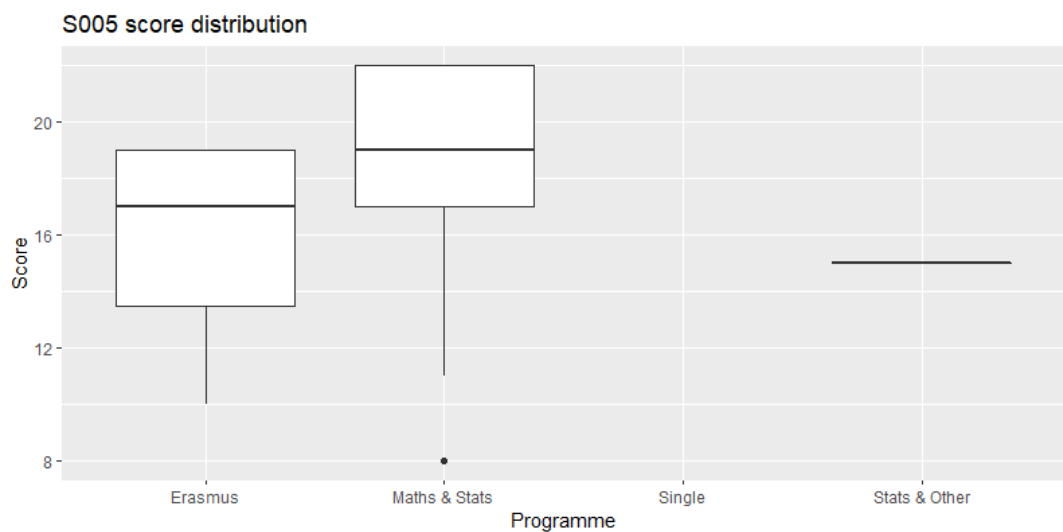


FIGURE B.15: Score distribution of S005 in each programme of study

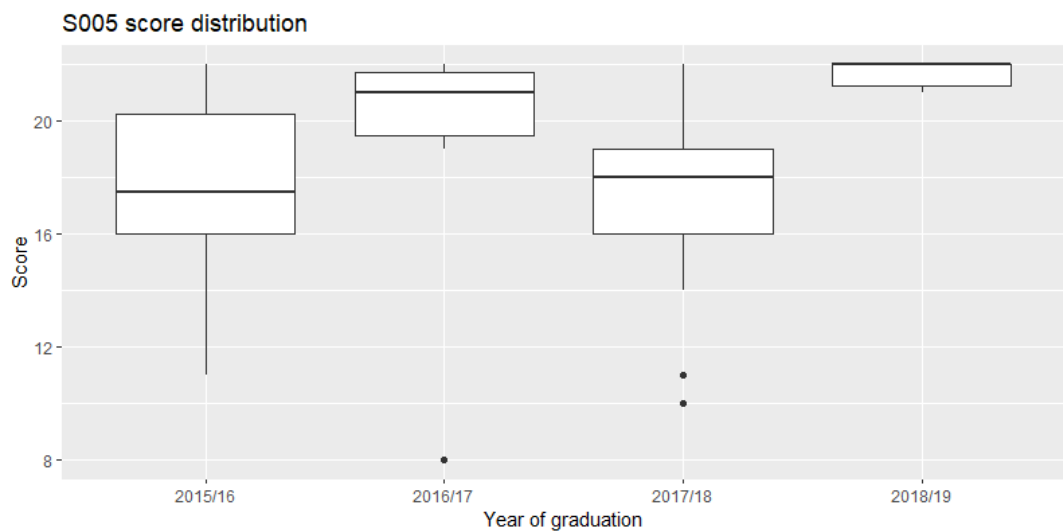


FIGURE B.16: Score distribution of S005 in each year of study

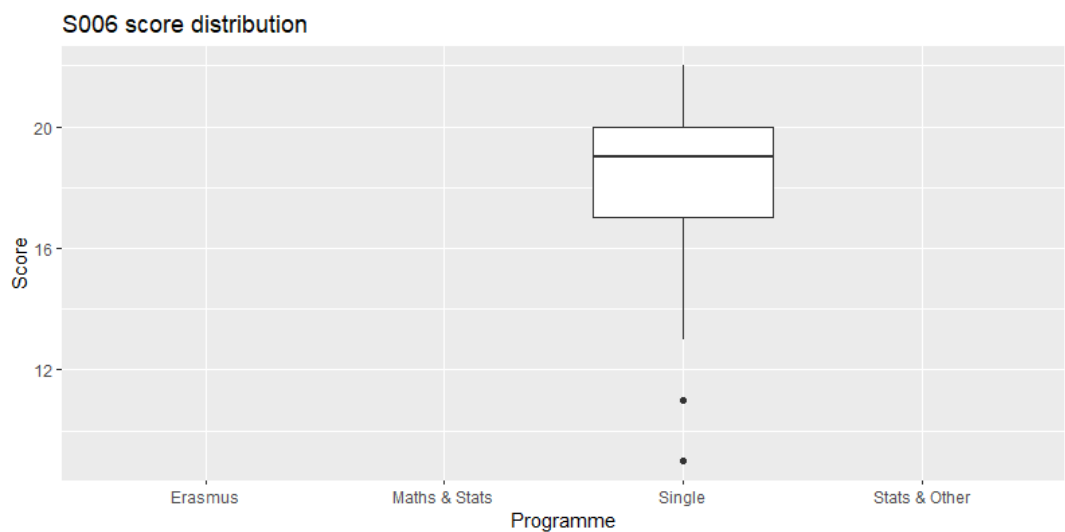


FIGURE B.17: Score distribution of S006 in each programme of study

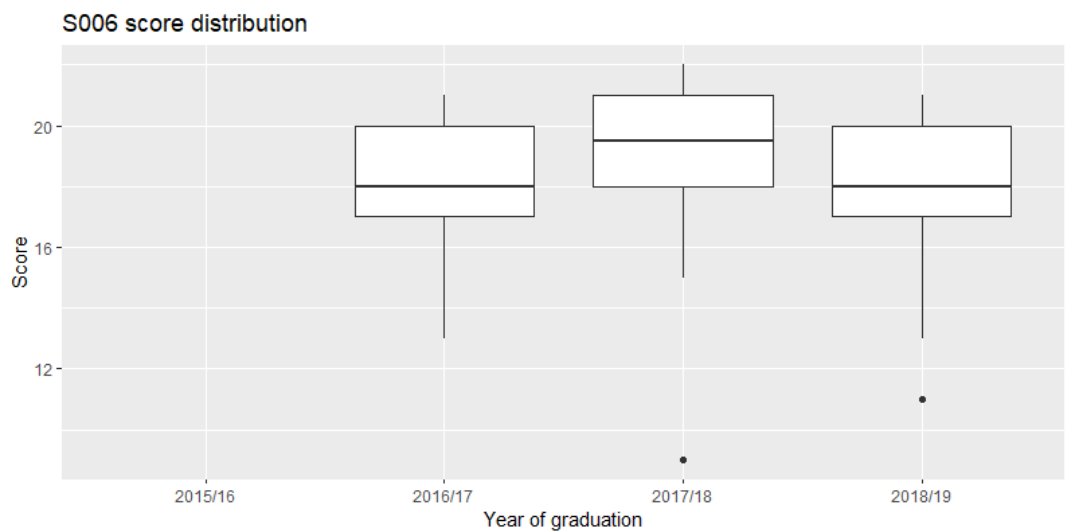


FIGURE B.18: Score distribution of S006 in each year of study

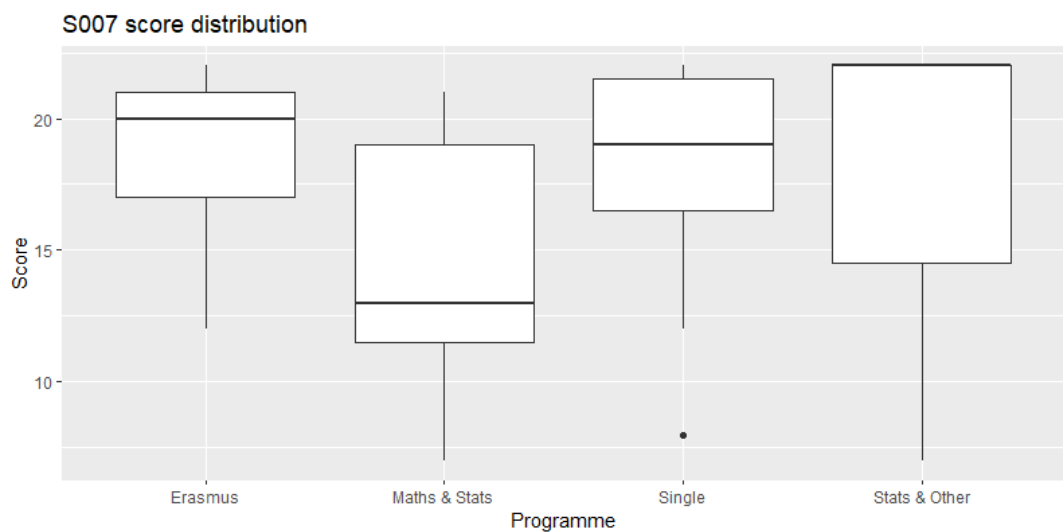


FIGURE B.19: Score distribution of S007 in each programme of study

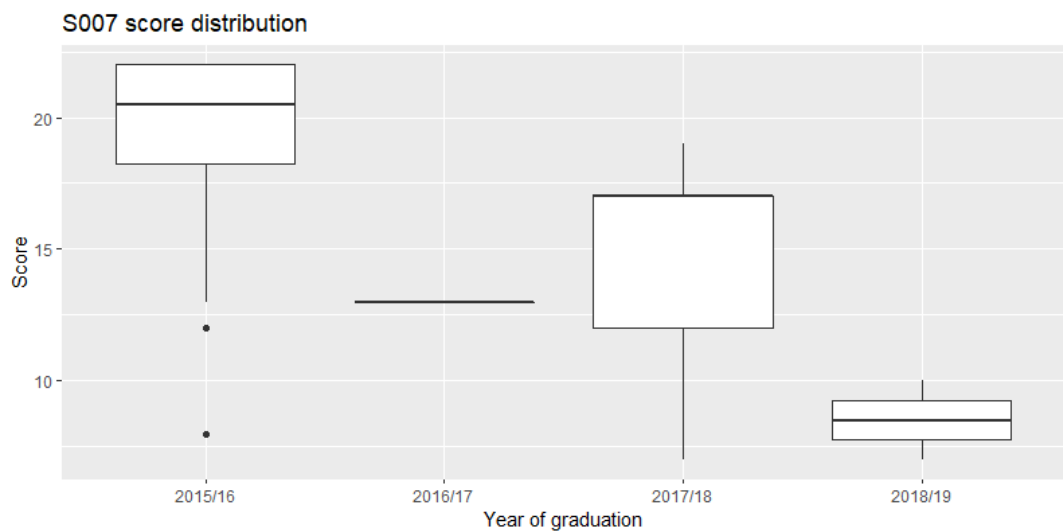


FIGURE B.20: Score distribution of S007 in each year of study

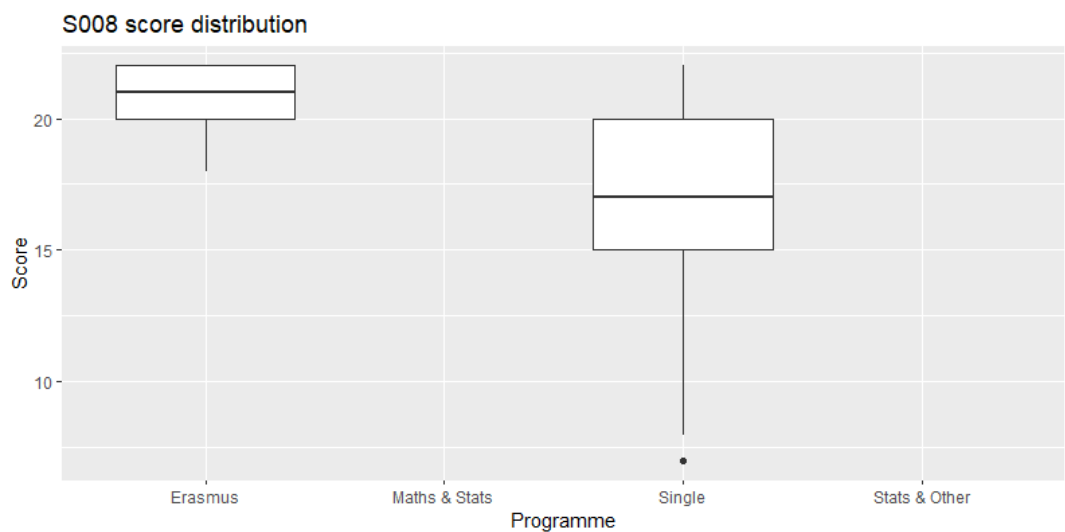


FIGURE B.21: Score distribution of S008 in each programme of study

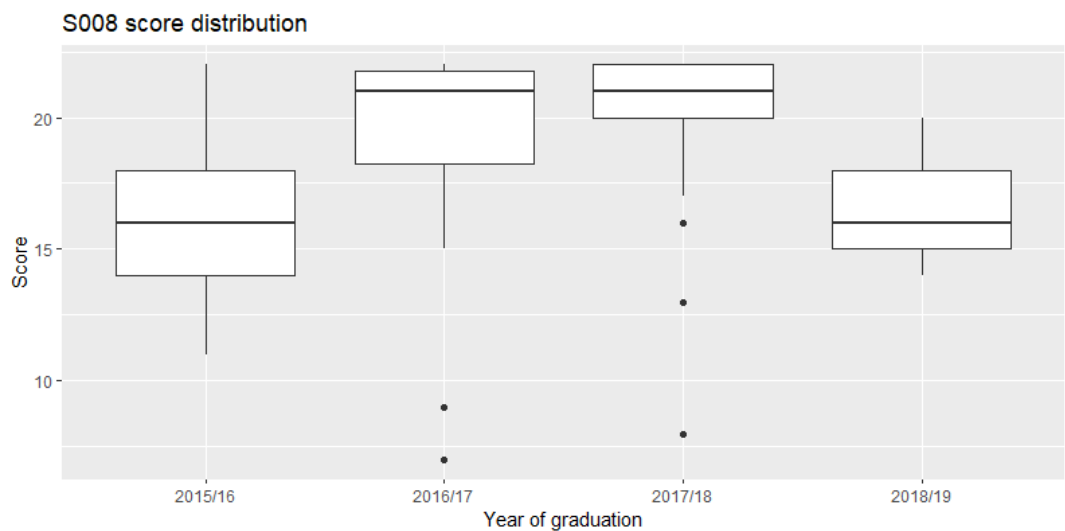


FIGURE B.22: Score distribution of S008 in each year of study

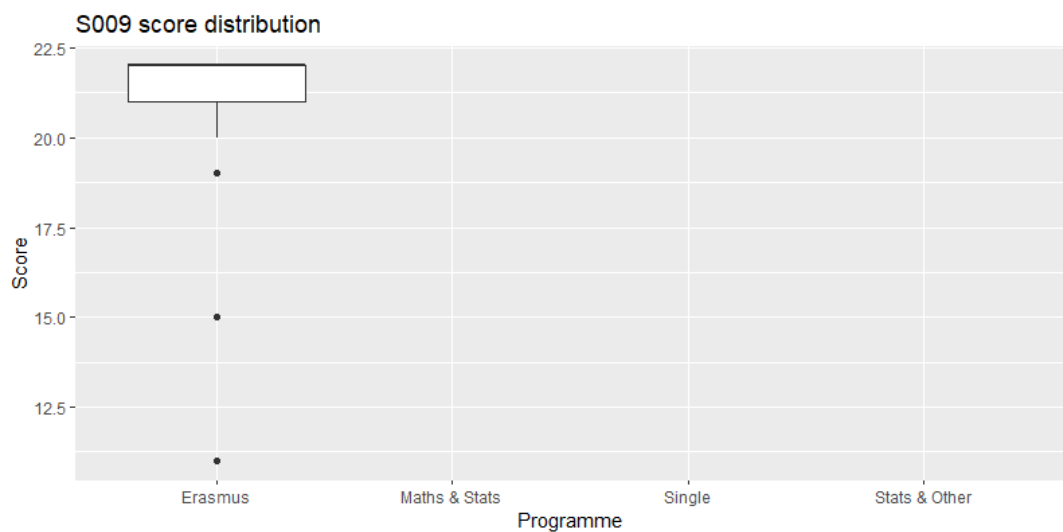


FIGURE B.23: Score distribution of S009 in each programme of study

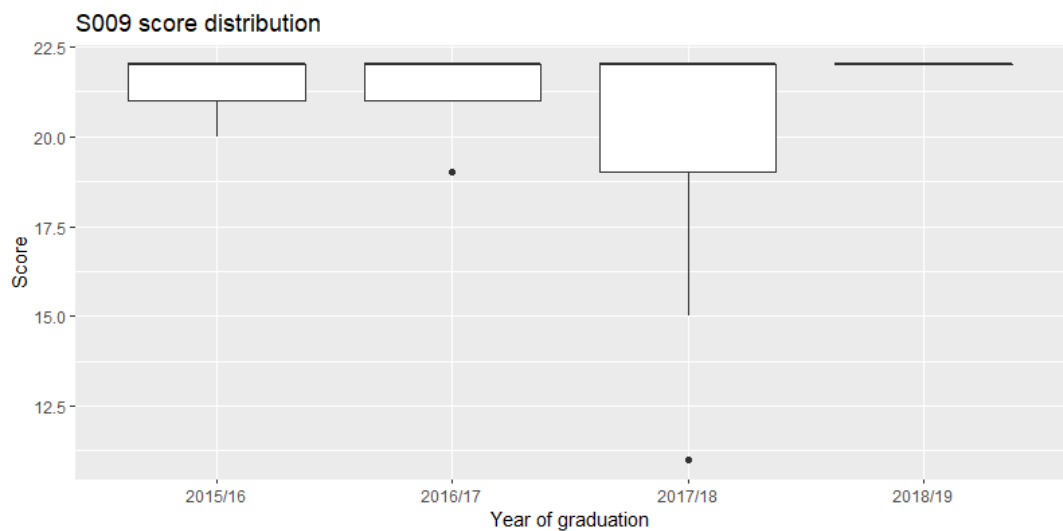


FIGURE B.24: Score distribution of S009 in each year of study

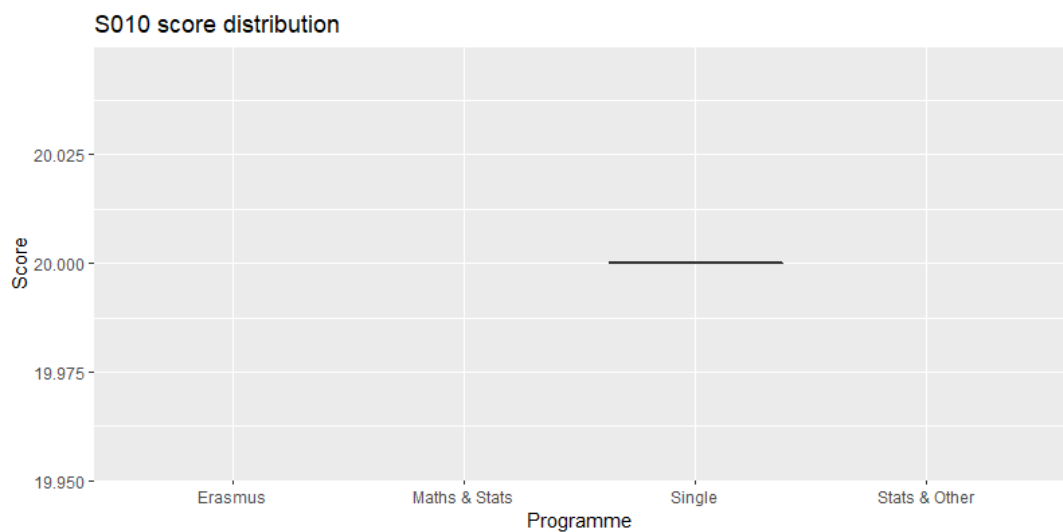


FIGURE B.25: Score distribution of S010 in each programme of study

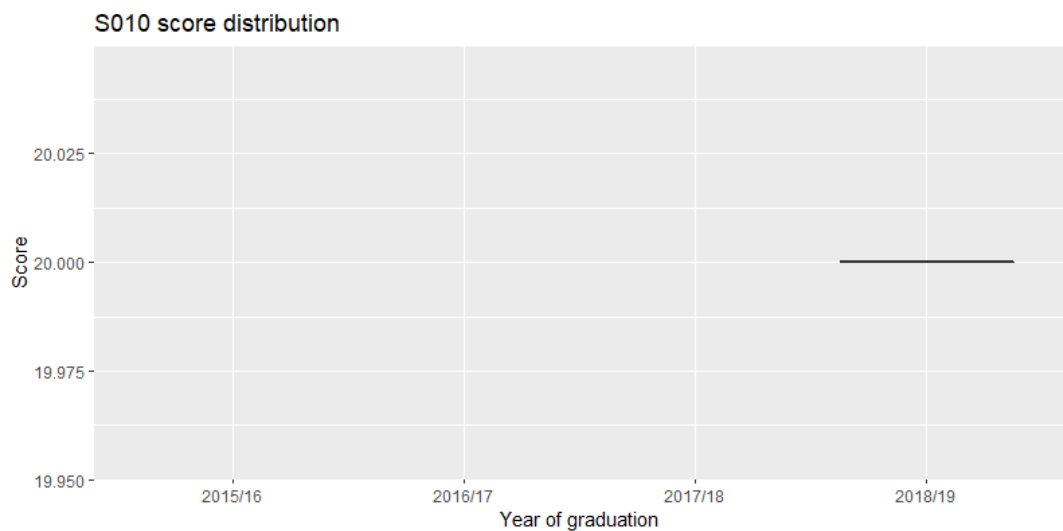


FIGURE B.26: Score distribution of S010 in each year of study

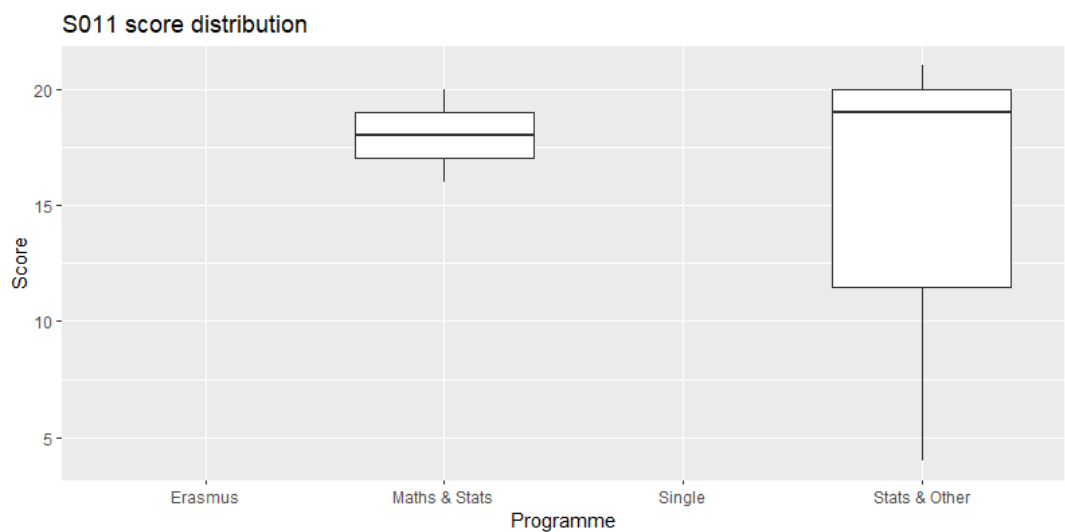


FIGURE B.27: Score distribution of S011 in each programme of study

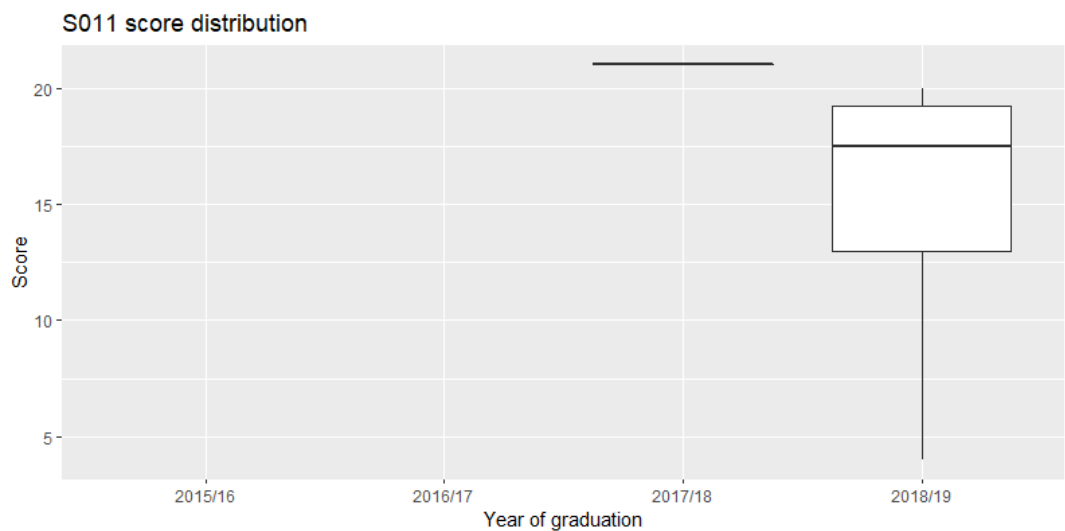


FIGURE B.28: Score distribution of S011 in each year of study

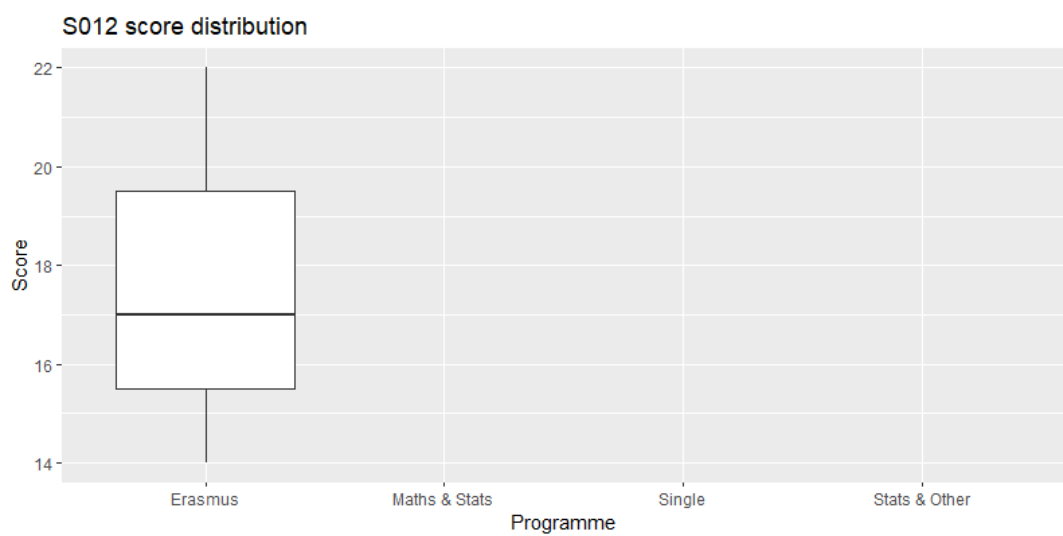


FIGURE B.29: Score distribution of S012 in each programme of study

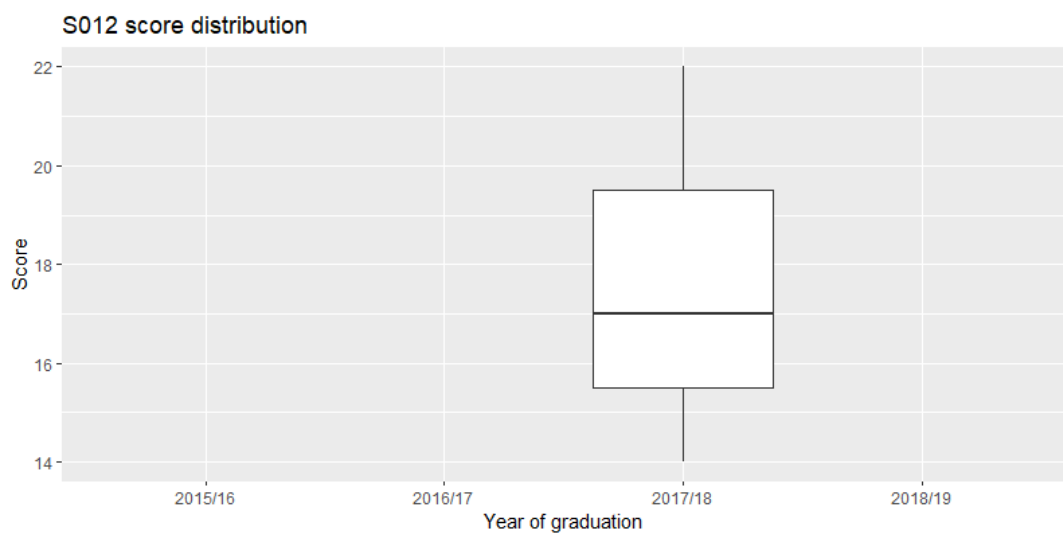


FIGURE B.30: Score distribution of S012 in each year of study

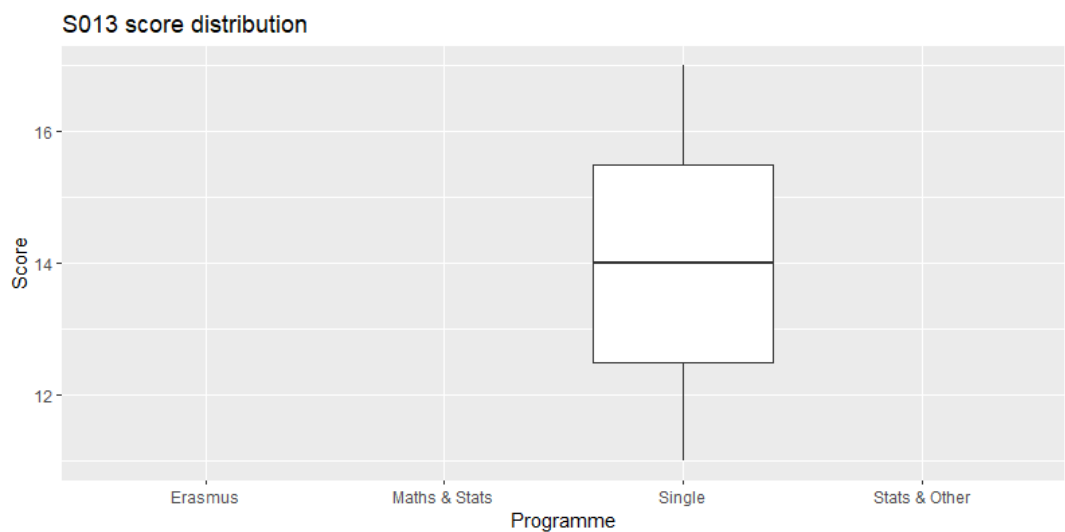


FIGURE B.31: Score distribution of S013 in each programme of study

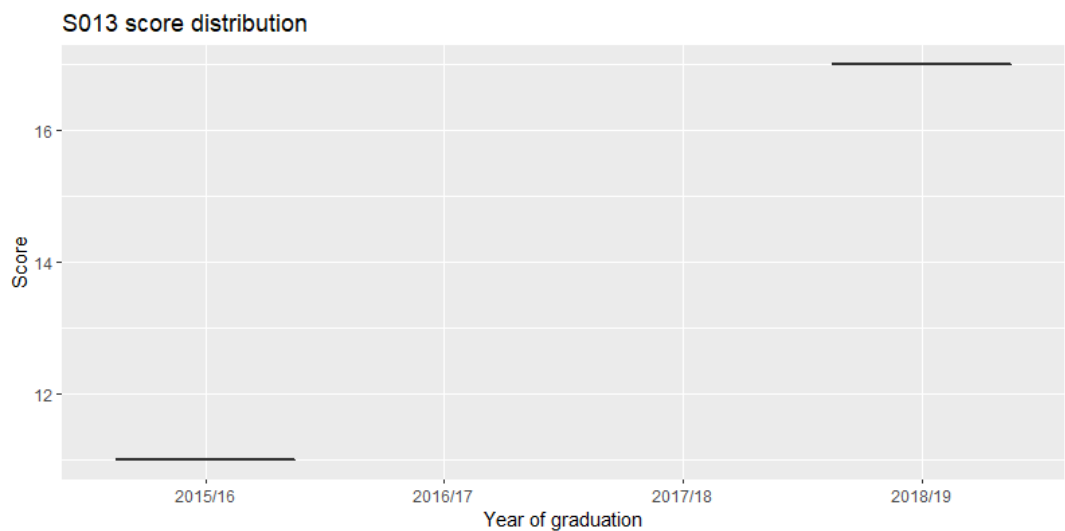


FIGURE B.32: Score distribution of S013 in each year of study

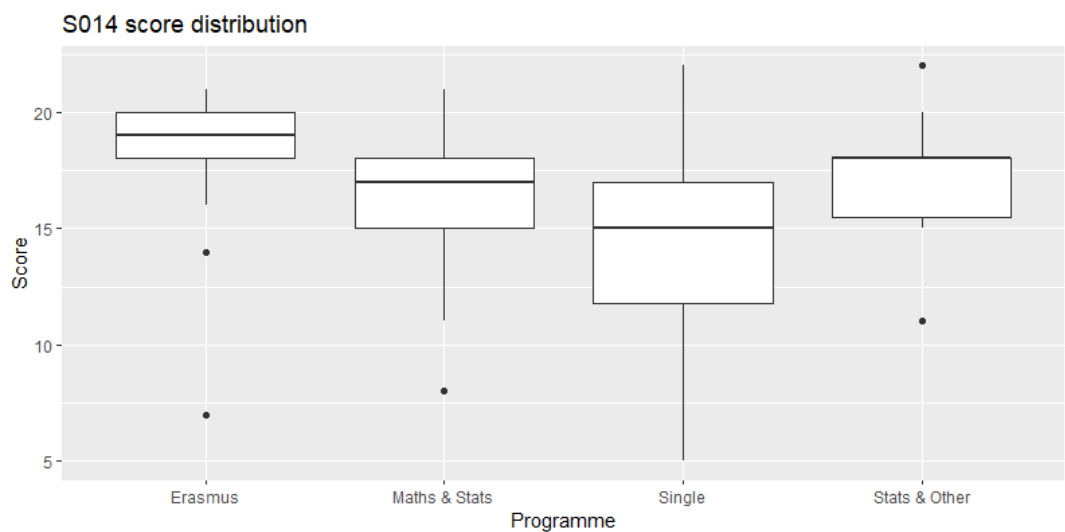


FIGURE B.33: Score distribution of S014 in each programme of study

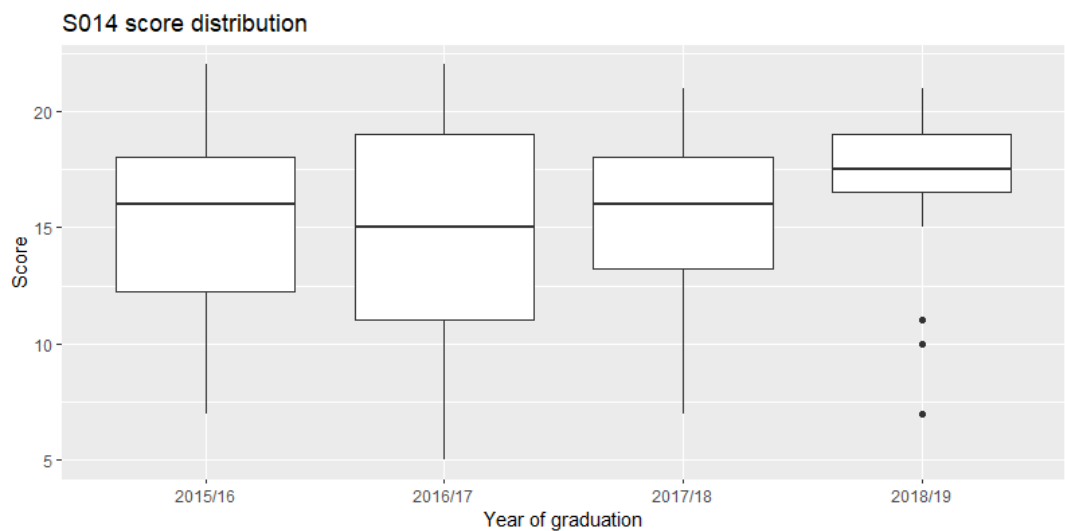


FIGURE B.34: Score distribution of S014 in each year of study

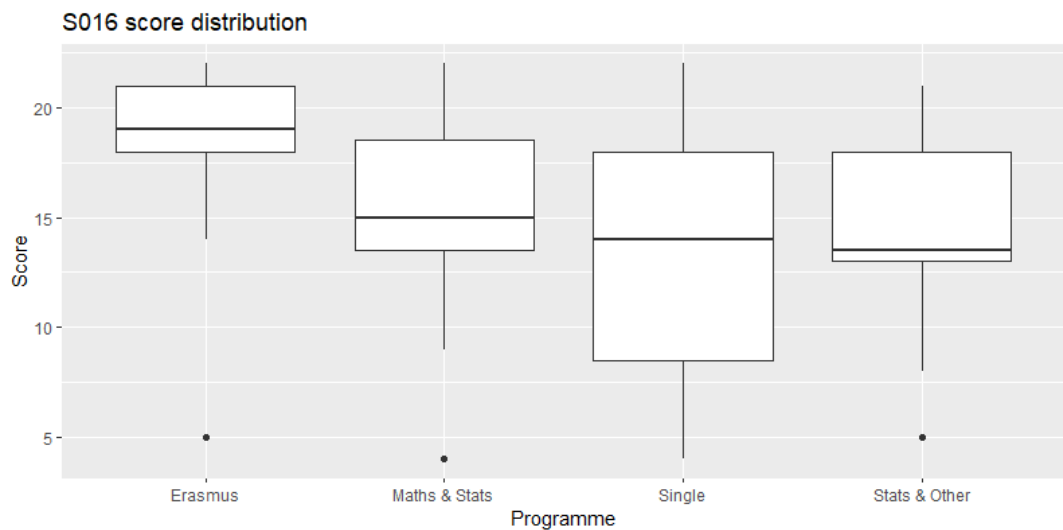


FIGURE B.35: Score distribution of S016 in each programme of study

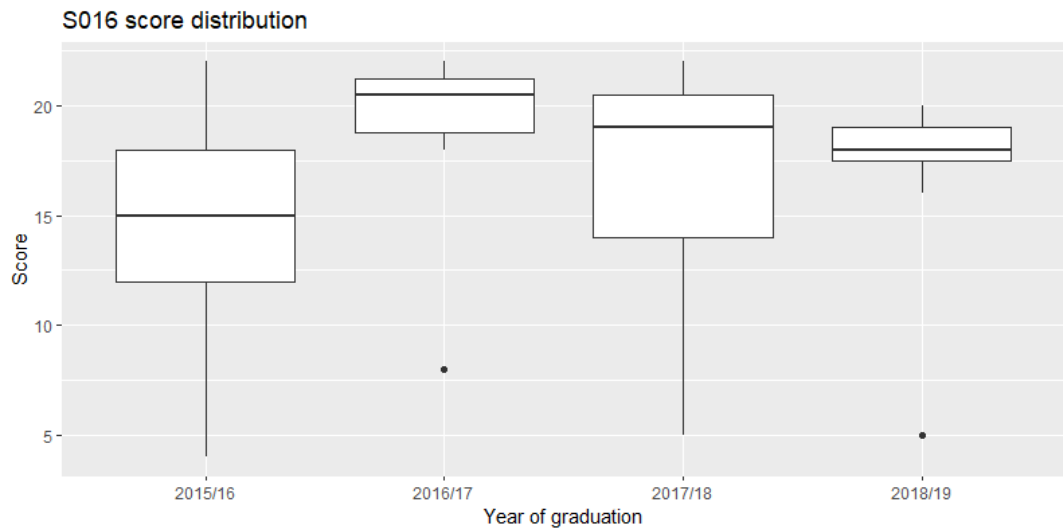


FIGURE B.36: Score distribution of S016 in each year of study

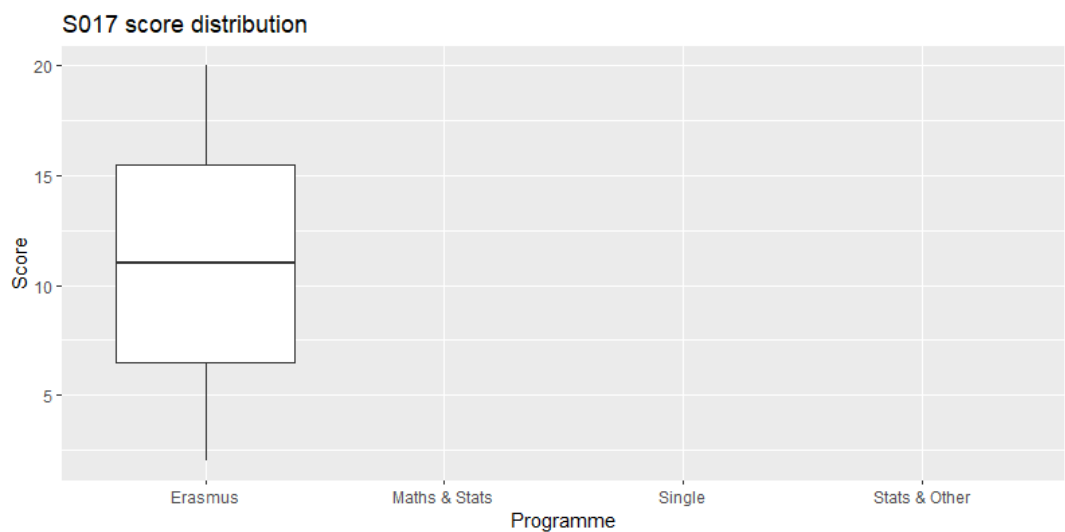


FIGURE B.37: Score distribution of S017 in each programme of study

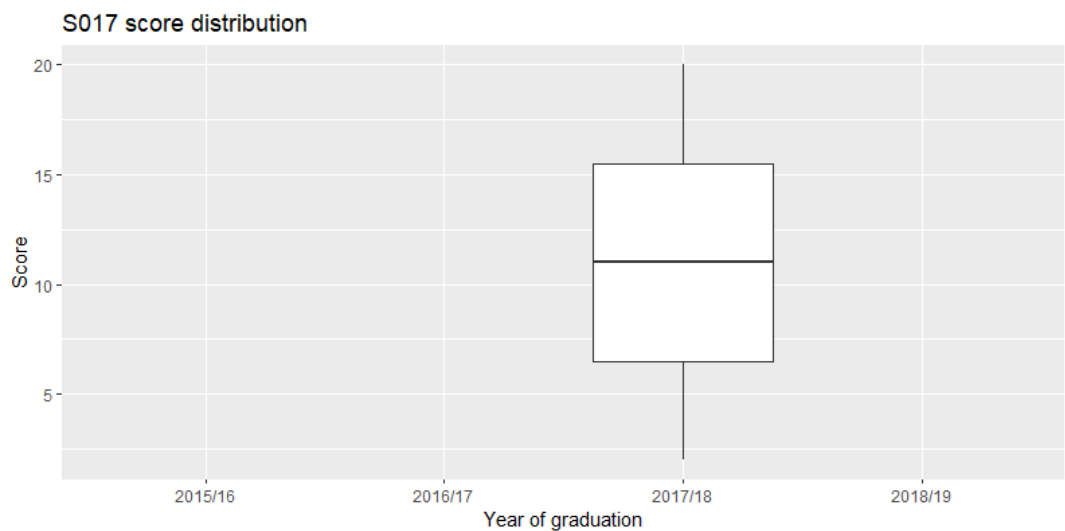


FIGURE B.38: Score distribution of S017 in each year of study

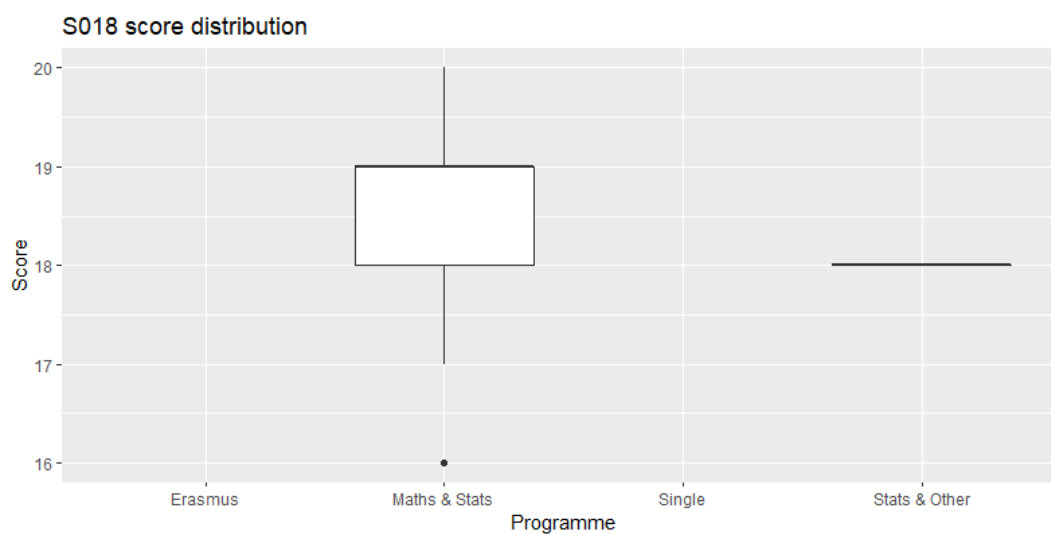


FIGURE B.39: Score distribution of S018 in each programme of study

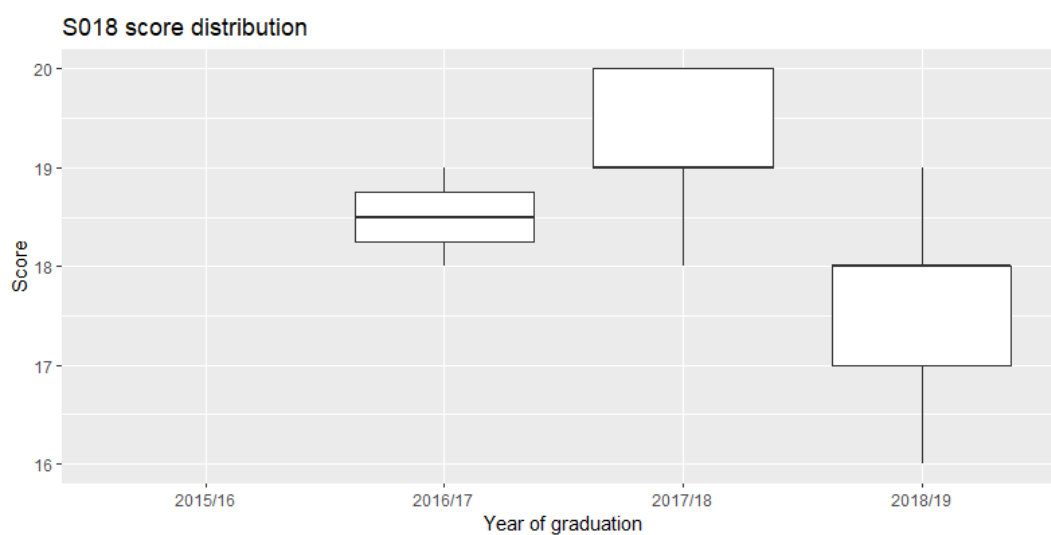


FIGURE B.40: Score distribution of S018 in each year of study

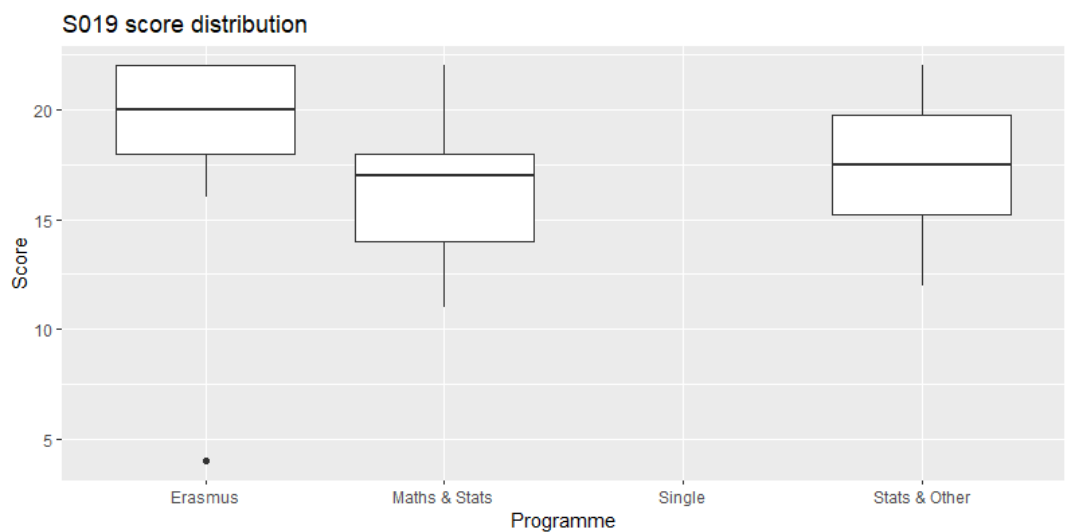


FIGURE B.41: Score distribution of S019 in each programme of study

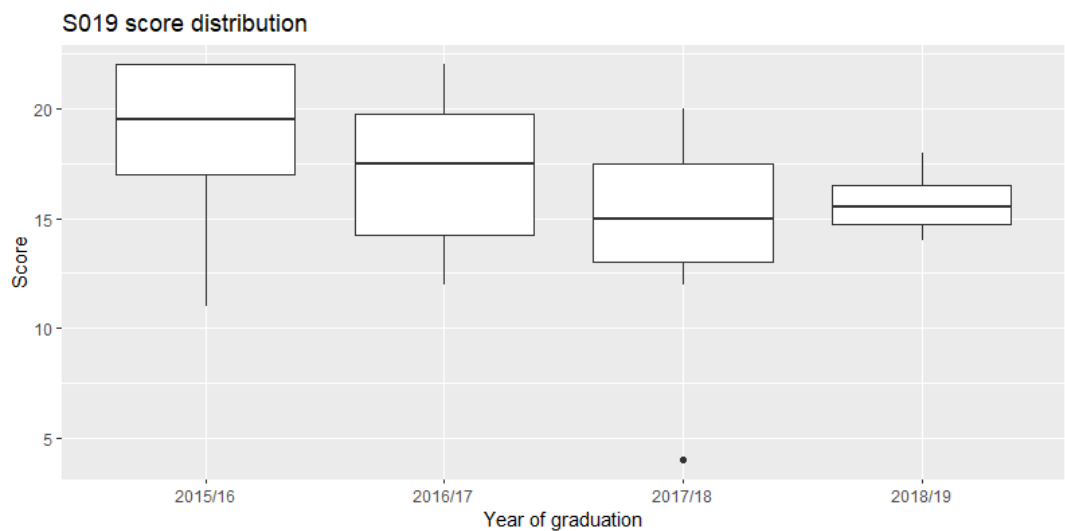


FIGURE B.42: Score distribution of S019 in each year of study

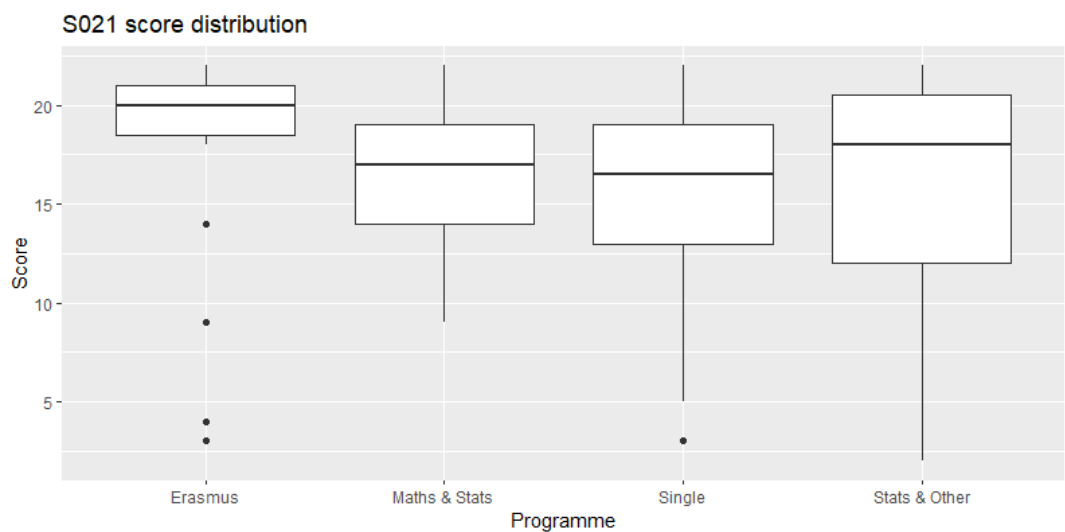


FIGURE B.43: Score distribution of S021 in each programme of study

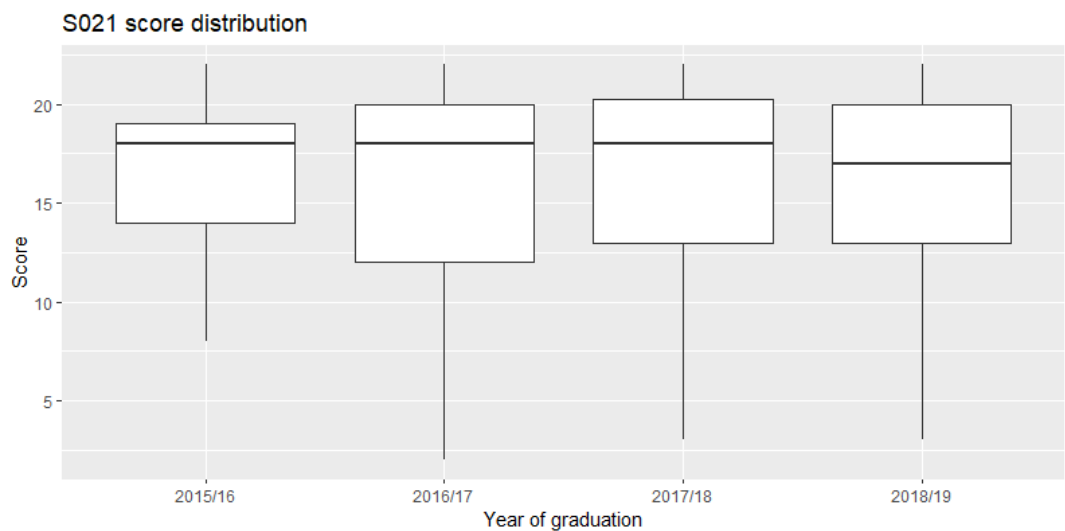


FIGURE B.44: Score distribution of S021 in each year of study

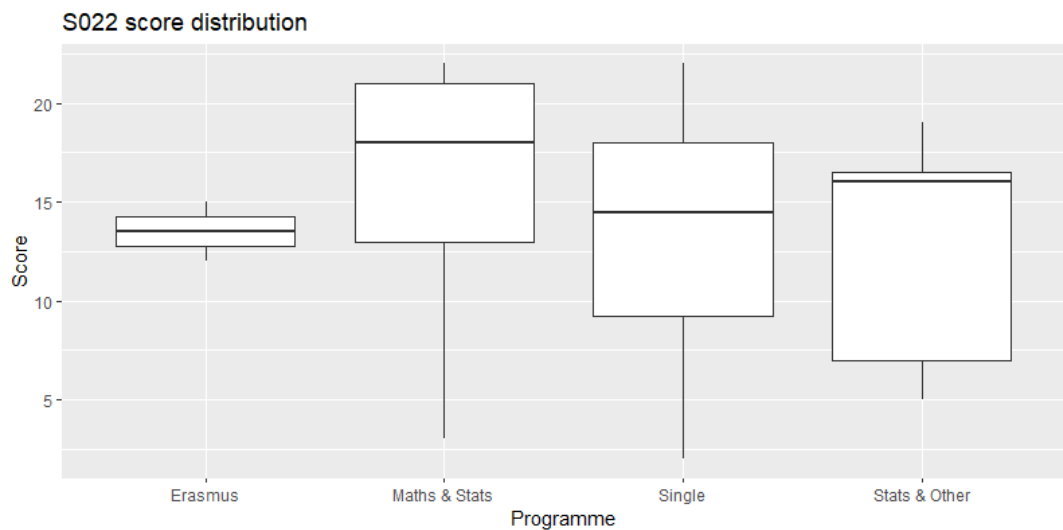


FIGURE B.45: Score distribution of S022 in each programme of study

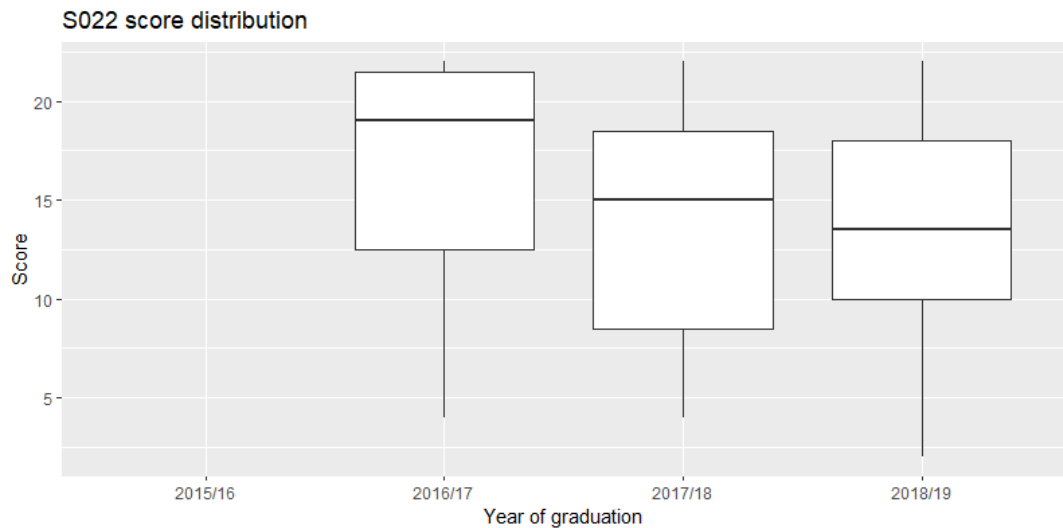


FIGURE B.46: Score distribution of S022 in each year of study

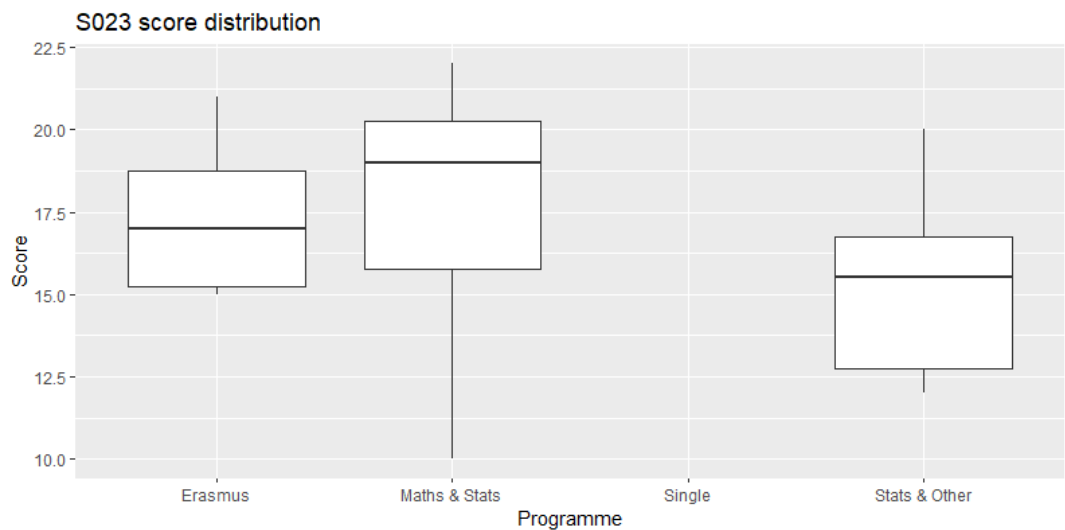


FIGURE B.47: Score distribution of S023 in each programme of study

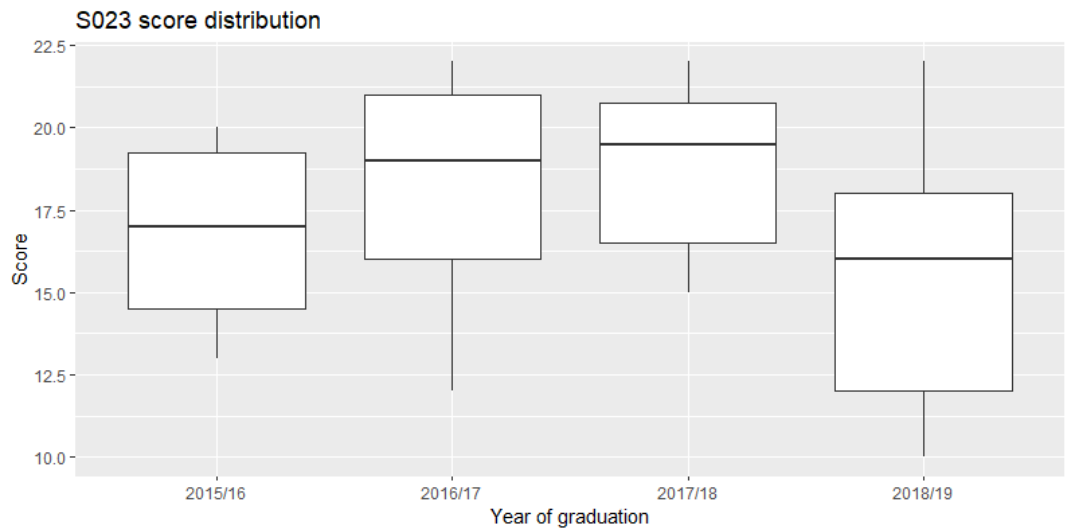


FIGURE B.48: Score distribution of S023 in each year of study

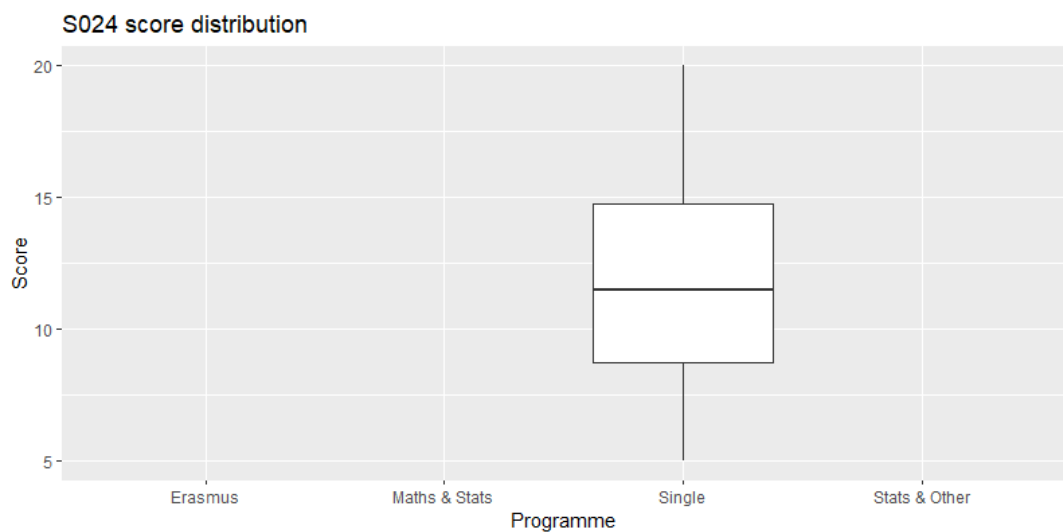


FIGURE B.49: Score distribution of S024 in each programme of study

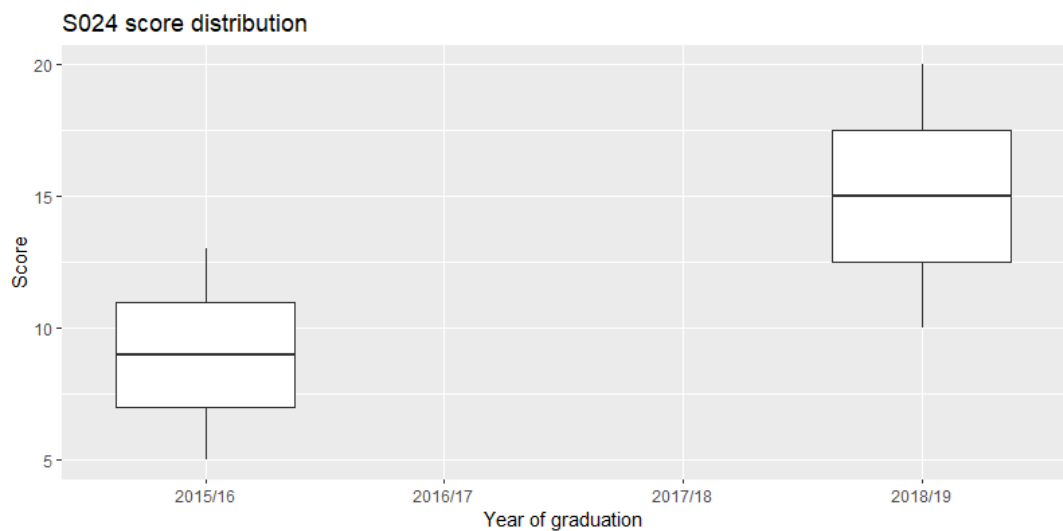


FIGURE B.50: Score distribution of S024 in each year of study

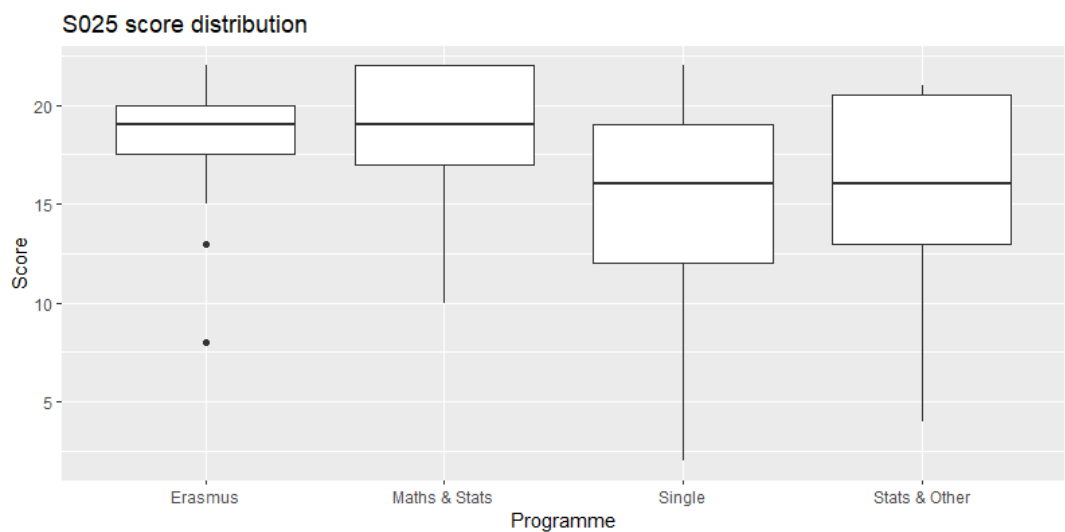


FIGURE B.51: Score distribution of S025 in each programme of study

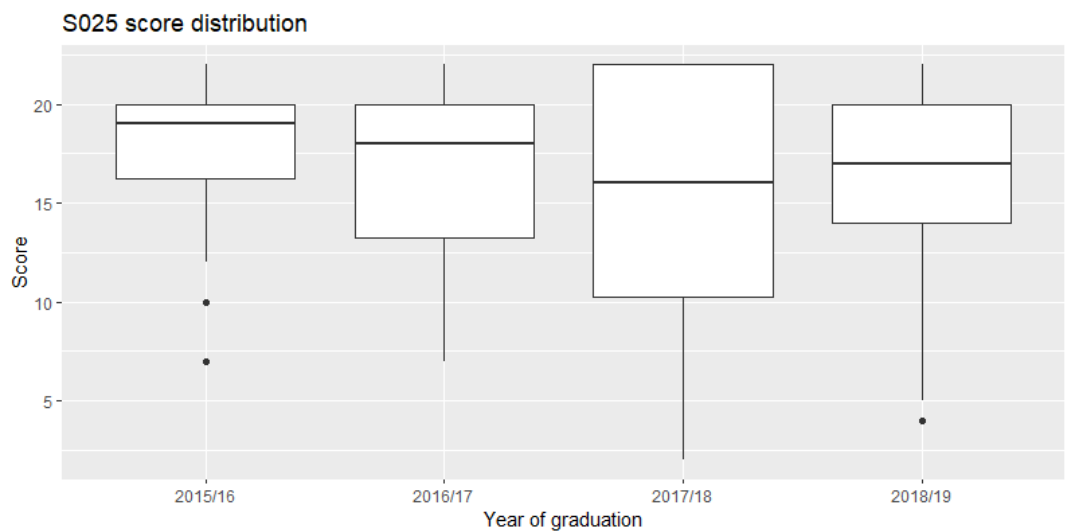


FIGURE B.52: Score distribution of S025 in each year of study

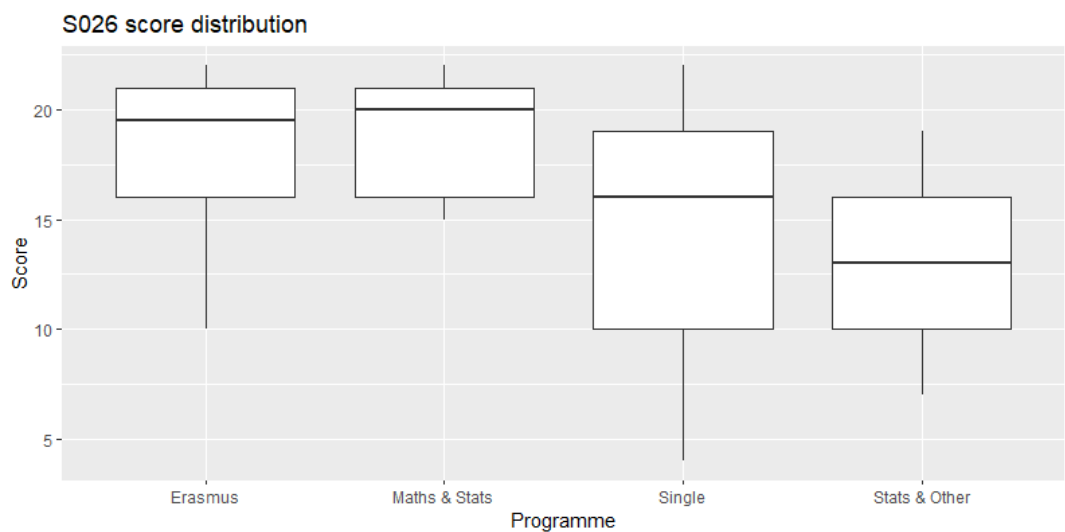


FIGURE B.53: Score distribution of S026 in each programme of study

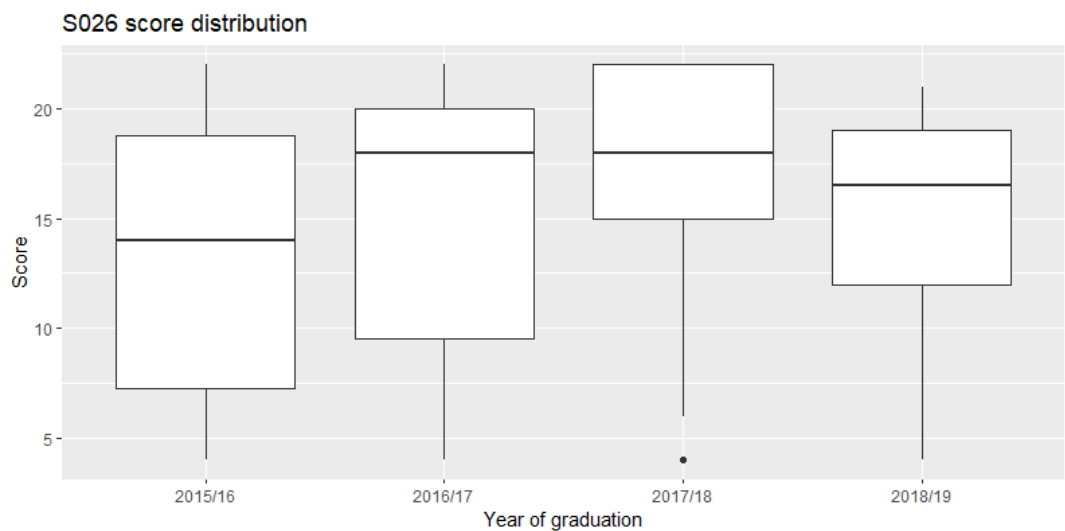


FIGURE B.54: Score distribution of S026 in each year of study

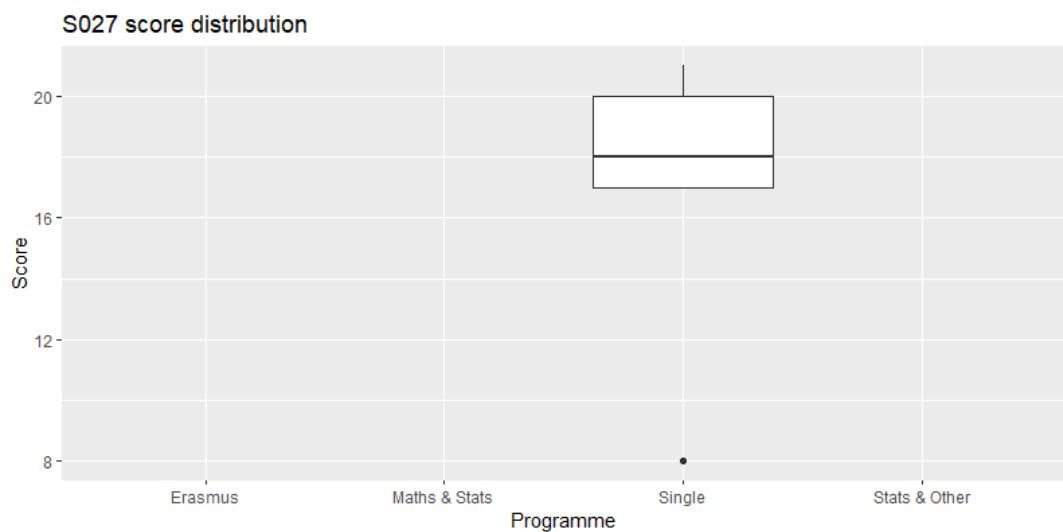


FIGURE B.55: Score distribution of S027 in each programme of study

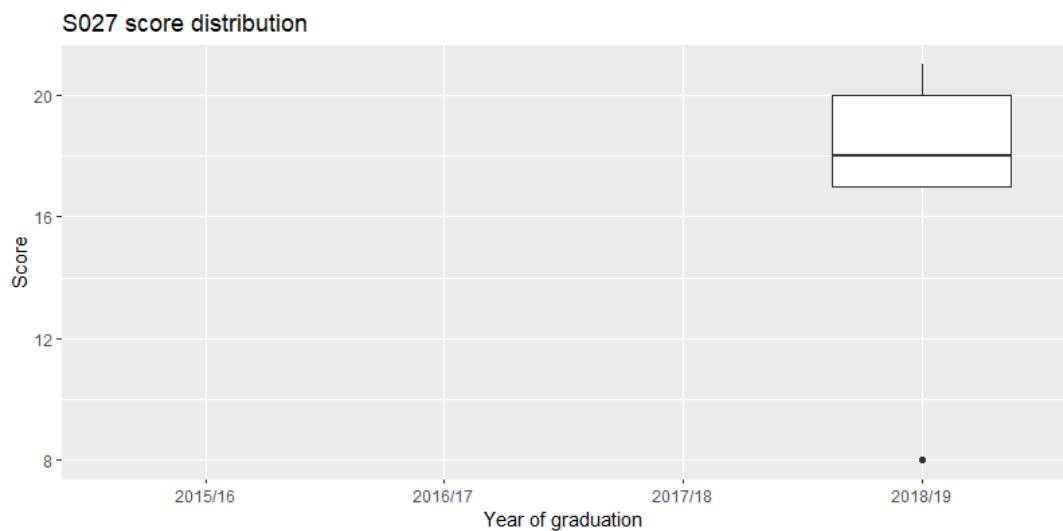


FIGURE B.56: Score distribution of S027 in each year of study

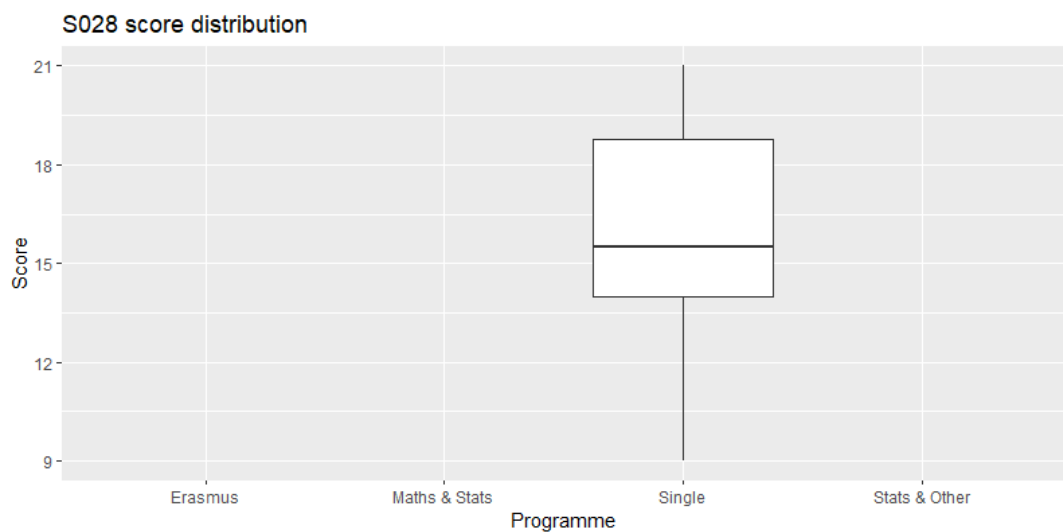


FIGURE B.57: Score distribution of S028 in each programme of study

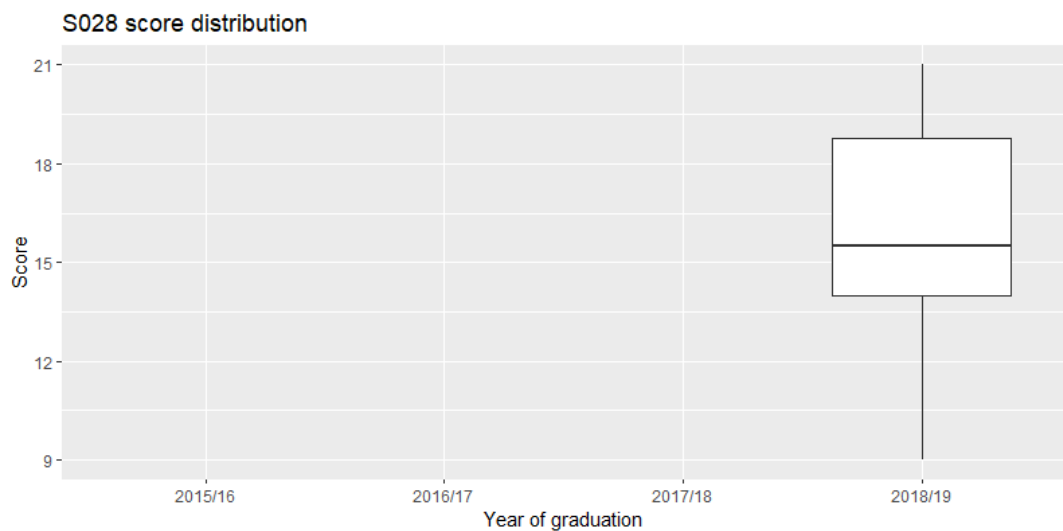


FIGURE B.58: Score distribution of S028 in each year of study

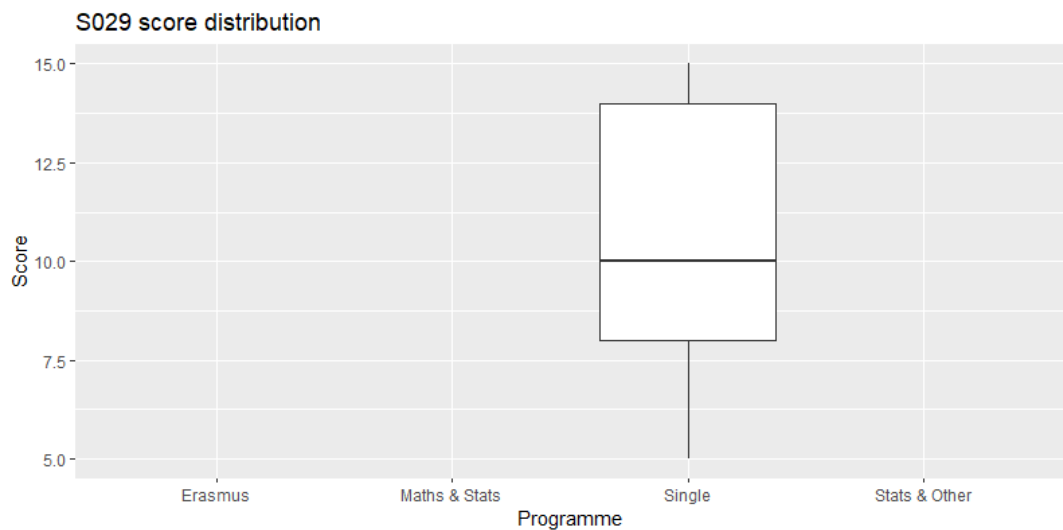


FIGURE B.59: Score distribution of S029 in each programme of study

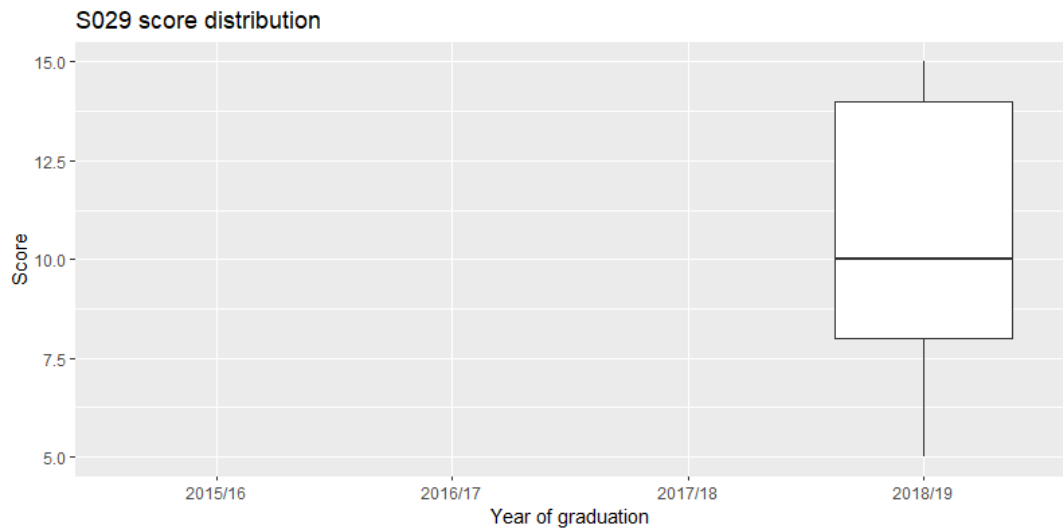


FIGURE B.60: Score distribution of S029 in each year of study

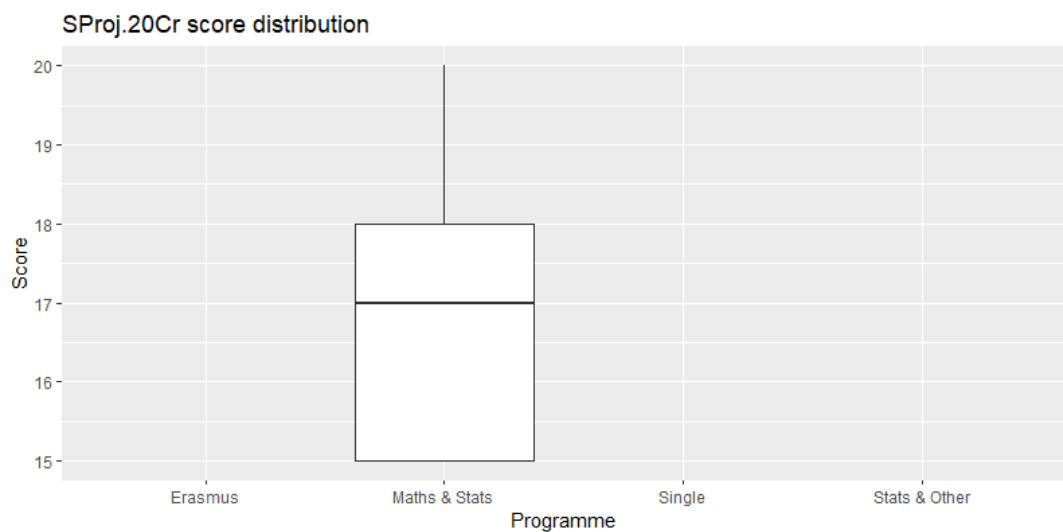


FIGURE B.61: Score distribution of SProj.20Cr in each programme of study

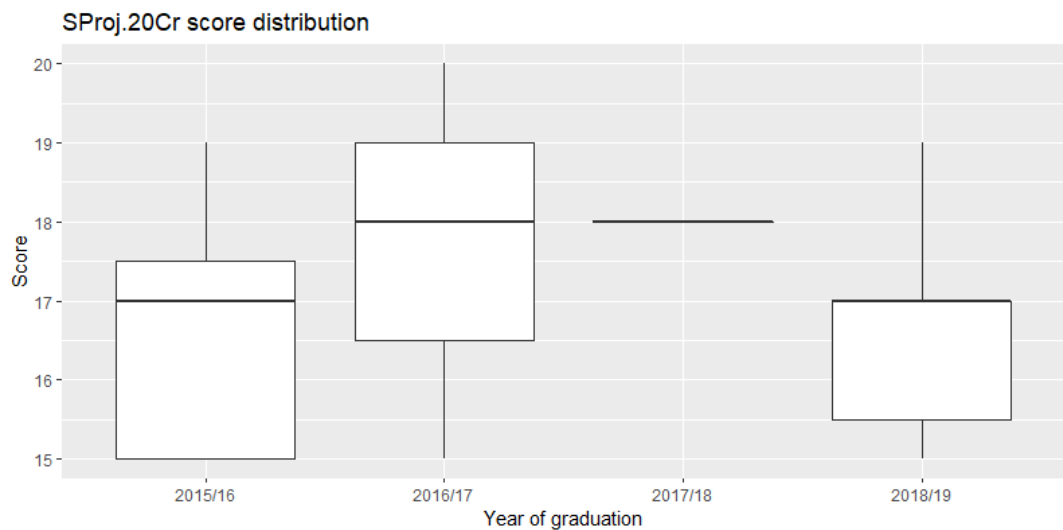


FIGURE B.62: Score distribution of SProj.20Cr in each year of study

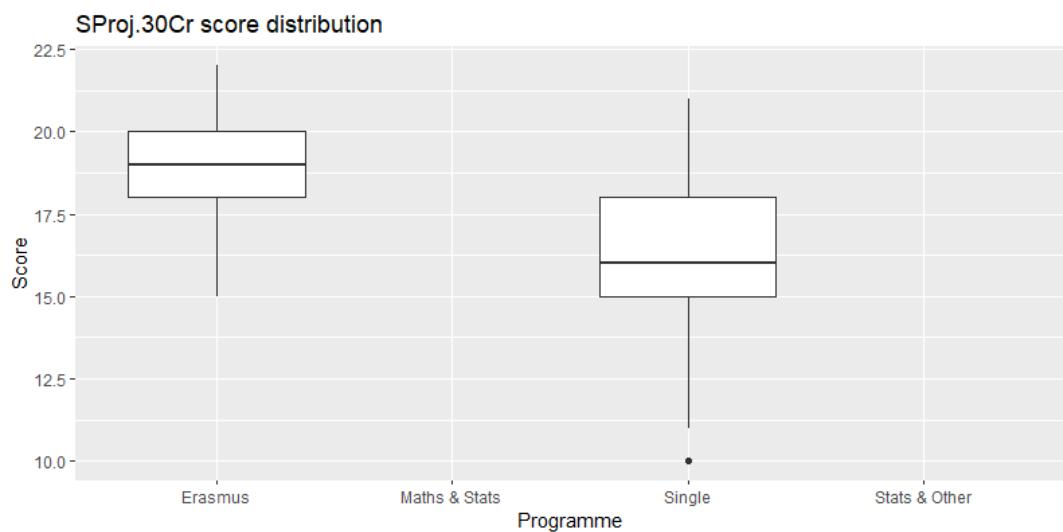


FIGURE B.63: Score distribution of SProj.30Cr in each programme of study

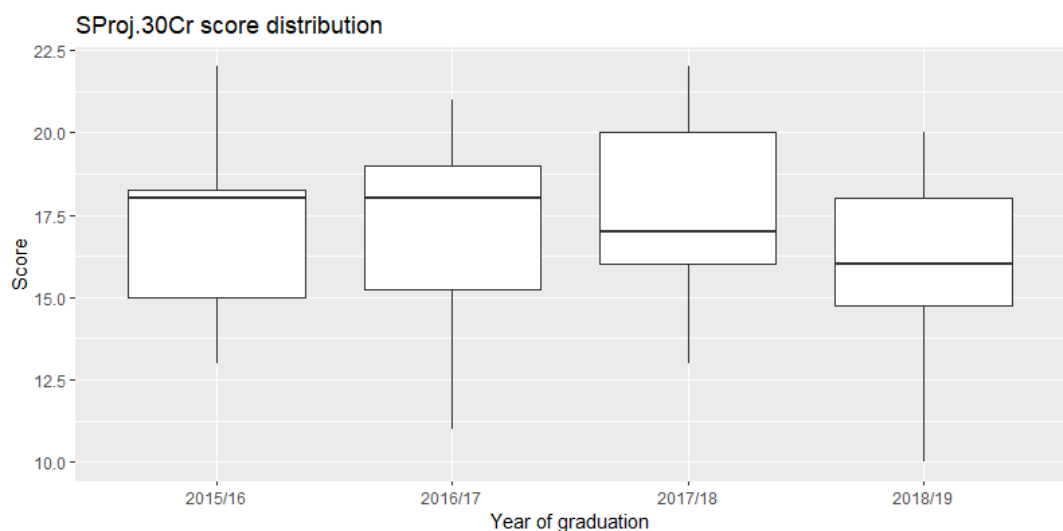


FIGURE B.64: Score distribution of SProj.30Cr in each year of study

Appendix C

Number of Students in Each Course

The total number of students is 190. There are 28 courses, four programmes of study and four years of graduation in total.

	S010	S003	S013	S012	S017	S024	S011	S029	S027	S028
Total	1	1	2	3	3	4	5	5	9	10
Erasmus	0	0	0	3	3	0	0	0	0	0
Math & Stats	0	0	0	0	0	0	2	0	0	0
Single	1	1	2	0	0	4	0	5	9	10
Stats & Other	0	0	0	0	0	0	3	0	0	0

	S018	S009	SProj.20Cr	S005	S023	S007	S019	S006	S016	S008
Total	12	24	25	39	40	42	47	68	76	110
Erasmus	0	24	0	7	6	9	11	0	24	20
Math & Stats	11	0	25	31	28	7	22	0	15	0
Single	0	0	0	0	0	23	0	68	23	90
Stats & Other	1	0	0	1	6	3	14	0	14	0

	S022	SProj.30Cr	S002	S014	S026	S001	S025	S021
Total	110	111	117	117	118	137	146	150
Erasmus	2	20	13	22	20	22	22	27
Math & Stats	35	0	8	13	7	14	25	18
Single	66	91	87	72	89	89	80	90
Stats & Other	7	0	9	10	2	12	19	15

TABLE C.1: Number of students by programme of study

	S010	S003	S013	S012	S017	S024	S011	S029	S027	S028
Total	1	1	2	3	3	4	5	5	9	10
2015/16	0	1	1	0	0	2	0	0	0	0
2016/17	0	0	0	0	0	0	0	0	0	0
2017/18	0	0	0	3	3	0	1	0	0	0
2018/19	1	0	1	0	0	2	4	5	9	10

	S018	S009	SProj.20Cr	S005	S023	S007	S019	S006	S016	S008
Total	12	24	25	39	40	42	47	68	76	110
2015/16	0	5	8	10	4	34	18	0	49	27
2016/17	2	5	8	6	9	1	18	17	8	22
2017/18	5	9	2	17	14	5	7	24	11	29
2018/19	5	5	7	6	13	2	4	27	8	32

	S022	SProj.30Cr	S002	S014	S026	S001	S025	S021
Total	110	111	117	117	118	137	146	150
2015/16	0	28	27	34	26	42	38	45
2016/17	31	22	22	25	23	29	30	29
2017/18	39	29	34	38	37	33	42	40
2018/19	40	32	34	20	32	33	36	36

TABLE C.2: Number of students by graduation year

Appendix D

Model Assumption

This appendix shows the assumptions of linear models. In each figure, four assumptions are tested:

1. Residuals vs Fitted on top-left. We test the assumption that the residual mean is zero. A good indicator is a red horizontal line on zero without patterns.
2. Normal Q-Q on top-right. We test the assumption of residual normality. A good indicator is residuals are on a dashed line.
3. Scale-Location on bottom-left. We test the homogeneity of variance of the residuals. A good indicator is a red non-tilted horizontal line with residuals spread equally.
4. Residuals vs Leverage on bottom-right. We test if there exist extreme observations that are influential to the model result. A good indicator is that no points lie beyond the Cooks distance, the red dashed line.

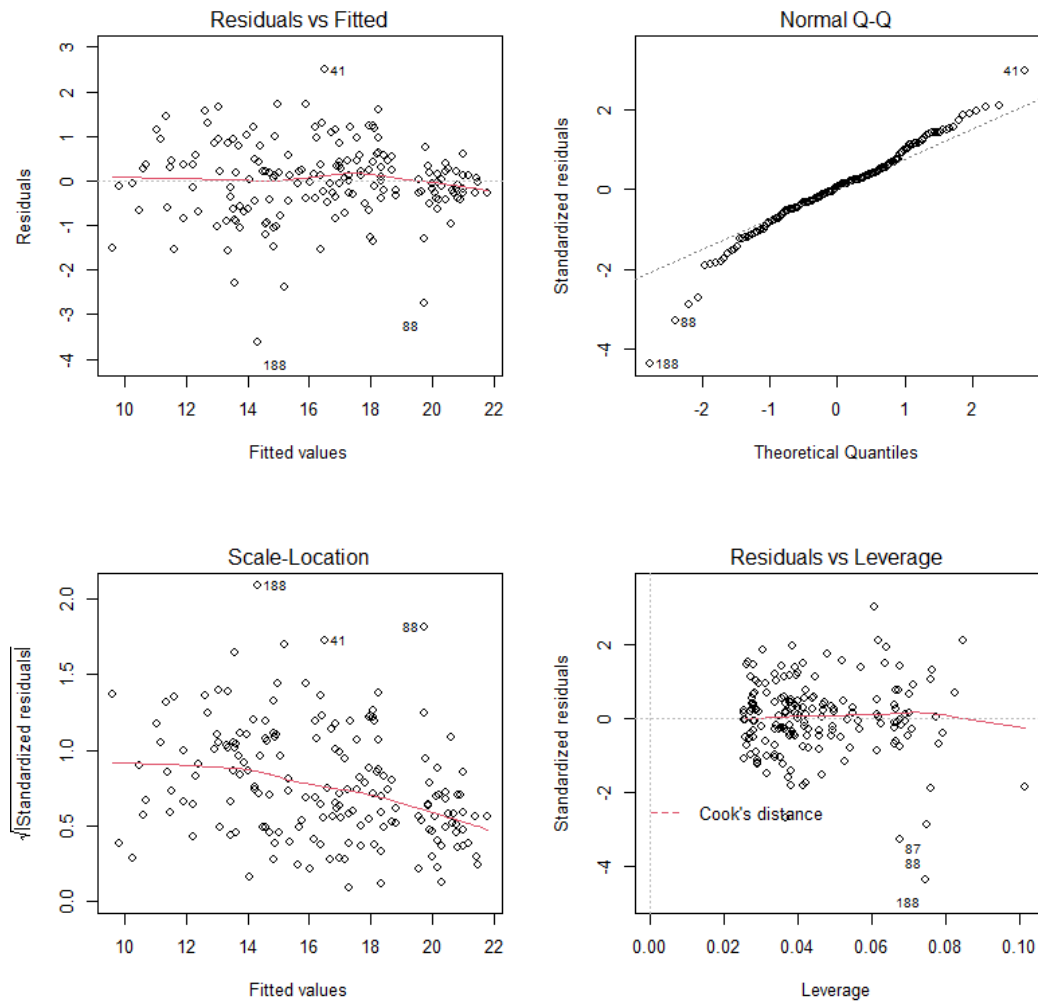


FIGURE D.1: Residual plot for a model
 $\text{Overall_aggregate} \sim \text{Level3_aggregate} + \text{Programme} + \text{Year}$

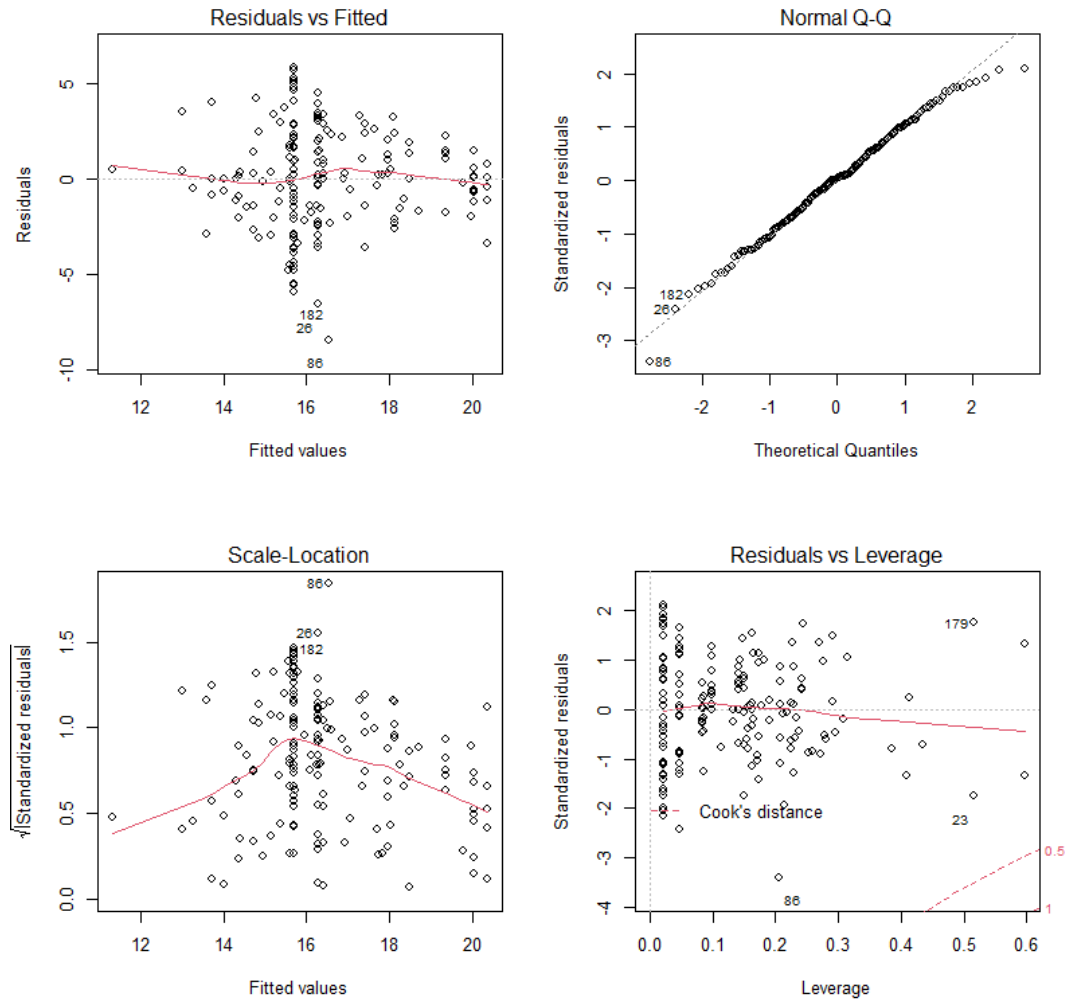


FIGURE D.2: Residual plot for a model
 $Overall_aggregate \sim \sum Course$

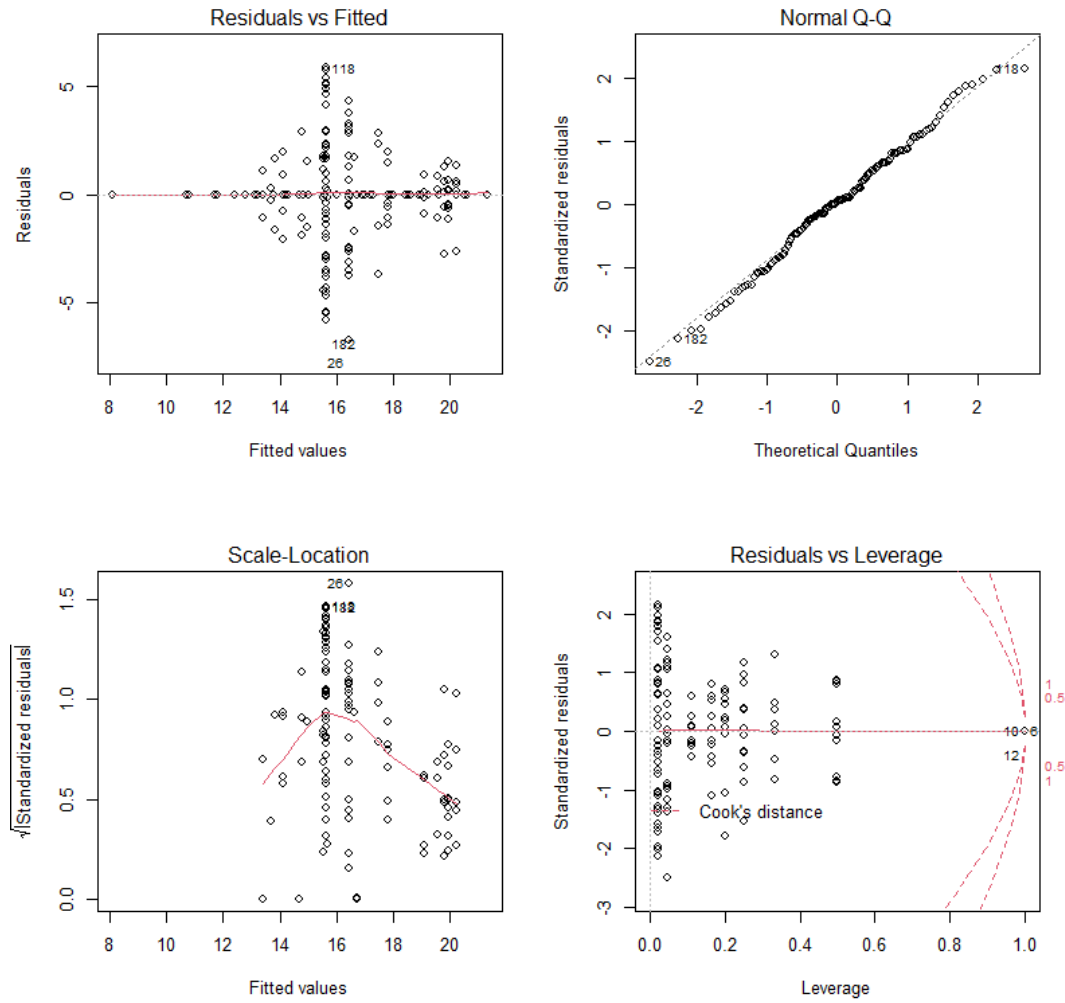


FIGURE D.3: Residual plot for a model
 $\text{Overall_aggregate} \sim \sum \text{Course} \times \sum \text{Course}$

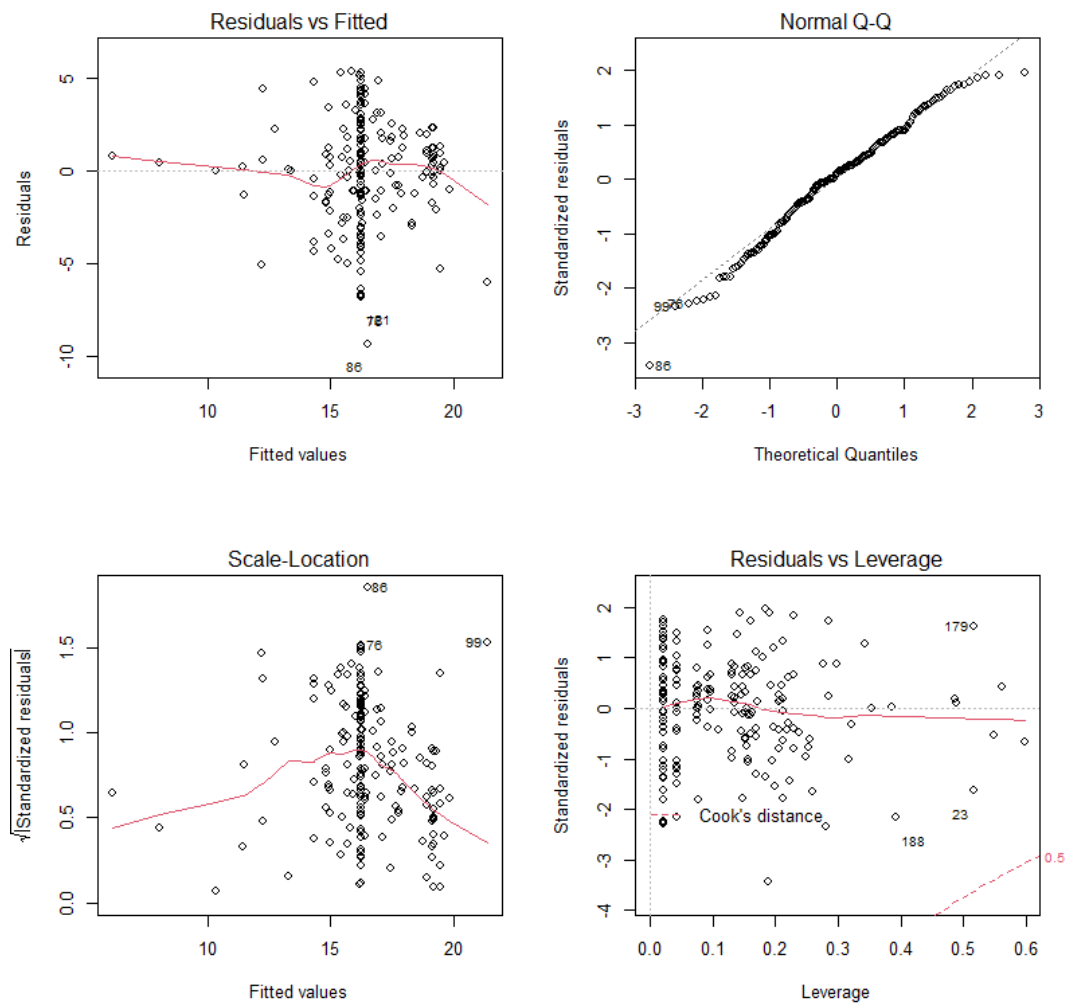


FIGURE D.4: Residual plot for a model
 $\text{Level4_aggregate} \sim \sum \text{Course}$

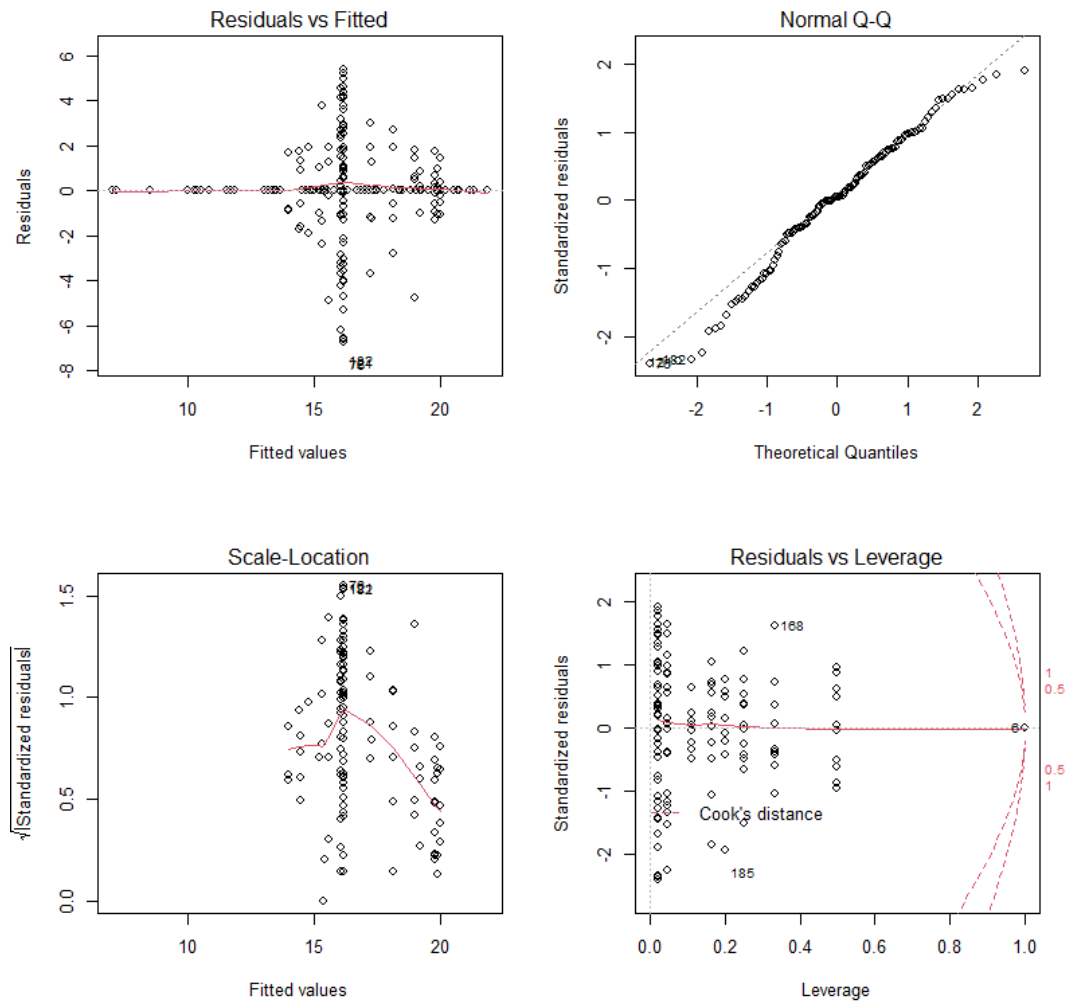


FIGURE D.5: Residual plot for a model
 $\text{Level4_aggregate} \sim \sum \text{Course} \times \sum \text{Course}$

Appendix E

Model Coefficients

In this appendix, the estimated values of each model variable are given. We do not show the variables that we cannot find an estimate theoretically.

Coefficients	Estimate	Std. Error	P-value
Intercept	0.5036	13.4454	0.9702
S026:TRUE	2.9791	2.2217	0.1819
S008:TRUE	1.8743	4.0109	0.6410
S007:TRUE	1.9658	2.2207	0.3774
S002:TRUE	2.2870	2.2075	0.3018
S023:TRUE	1.1351	2.2560	0.6156
S019:TRUE	2.6059	2.2777	0.2544
S014:TRUE	2.9103	2.3105	0.2097
S025:TRUE	2.7752	2.4599	0.2610
S021:TRUE	1.0061	2.2359	0.6534
S009:TRUE	5.7319	2.3224	0.0147
S001:TRUE	2.9608	2.4623	0.2311
S022:TRUE	3.3444	2.4390	0.1723
S018:TRUE	0.7449	2.6028	0.7751
S006:TRUE	1.7019	2.5042	0.4978
S016:TRUE	3.6548	2.4589	0.1393
SProj.30Cr:TRUE	-6.6402	5.7537	0.2503
SProj.20Cr:TRUE	6.6534	4.5638	0.1469
S005:TRUE	3.7084	2.5770	0.1522
S010:TRUE	2.2036	3.6001	0.5414
S024:TRUE	2.6727	3.7520	0.4773
S013:TRUE	2.5231	4.1404	0.5432
S011:TRUE	0.6325	2.7897	0.8209
S027:TRUE	4.6263	2.5972	0.0769
S028:TRUE	0.9418	2.4587	0.7022
S029:TRUE	1.9207	2.6454	0.4689

TABLE E.1: Coefficients for
Overall_aggregate $\sim \sum$ Course

Coefficients	Estimate	Std. Error	P-value
(Intercept)	93.9017	139.4064	0.5020
S026:TRUE	-74.3313	69.3385	0.2861
S008:TRUE	-53.0325	60.2811	0.3809
S007:TRUE	-39.7439	45.2893	0.3821
S002:TRUE	22.2033	42.0161	0.5983
S023:TRUE	-6.3922	35.7807	0.8585
S019:TRUE	-8.2839	26.9176	0.7589
S014:TRUE	-10.7833	24.0952	0.6554
S025:TRUE	-21.0378	23.4466	0.3715
S021:TRUE	-17.4256	41.4786	0.6752
S009:TRUE	30.6914	28.2340	0.2794
S001:TRUE	-1.4128	18.6755	0.9398
S022:TRUE	-9.0833	21.9620	0.6800
S018:TRUE	-0.0967	38.9590	0.9980
S006:TRUE	-1.5551	17.8734	0.9308
S016:TRUE	-16.2128	21.8844	0.4604
SProj.30Cr:TRUE	20.0376	75.2898	0.7906
SProj.20Cr:TRUE	-25.5711	39.9348	0.5233
S005:TRUE	-13.7128	20.1543	0.4977
S010:TRUE	23.6215	19.4834	0.2280
S024:TRUE	-20.7141	35.8615	0.5647
S013:TRUE	-10.2124	35.6101	0.7748
S011:TRUE	-9.8128	21.0826	0.6425
S027:TRUE	-15.6391	35.4409	0.6599
S028:TRUE	-18.3474	35.3854	0.6051
S029:TRUE	-19.0141	35.5219	0.5935
S026:TRUE and S008:TRUE	47.2786	64.6637	0.4662
S026:TRUE and S007:TRUE	66.6564	38.9568	0.0899
S026:TRUE and S002:TRUE	12.4376	20.8070	0.5512
S026:TRUE and S023:TRUE	-20.3004	25.3725	0.4254
S026:TRUE and S019:TRUE	9.1843	26.2428	0.7270
S026:TRUE and S014:TRUE	5.6187	31.1642	0.8573

TABLE E.2: Coefficients for
Overall_aggregate $\sim \sum \text{Course} \times \sum \text{Course}$

Coefficients	Estimate	Std. Error	P-value
S026:TRUE and S025:TRUE	15.5587	26.6660	0.5608
S026:TRUE and S021:TRUE	17.3776	21.7163	0.4253
S026:TRUE and S001:TRUE	-21.5104	11.5594	0.0654
S026:TRUE and S022:TRUE	52.2759	38.0961	0.1728
S026:TRUE and S018:TRUE	15.5700	35.6161	0.6629
S008:TRUE and S019:TRUE	5.1106	30.1583	0.8657
S007:TRUE and S002:TRUE	-36.1175	23.7255	0.1308
S007:TRUE and S023:TRUE	11.9697	41.1917	0.7719
S007:TRUE and S019:TRUE	2.9667	27.7673	0.9151
S007:TRUE and S014:TRUE	-6.7038	19.9640	0.7377
S007:TRUE and S025:TRUE	7.8697	20.6401	0.7037
S007:TRUE and S001:TRUE	35.2242	14.8111	0.0191
S007:TRUE and S022:TRUE	25.8903	18.9310	0.1742
S002:TRUE and S023:TRUE	-21.2750	12.0344	0.0799
S002:TRUE and S019:TRUE	10.3522	14.3978	0.4737
S002:TRUE and S014:TRUE	-2.1978	7.0957	0.7573
S002:TRUE and S025:TRUE	-10.9333	17.1313	0.5247
S002:TRUE and S021:TRUE	22.3917	27.0044	0.4088
S002:TRUE and S001:TRUE	-40.1311	19.7425	0.0445
S002:TRUE and S022:TRUE	-19.8956	11.5658	0.0882
S002:TRUE and S018:TRUE	-11.0872	25.4432	0.6639
S002:TRUE and S016:TRUE	5.7356	14.3534	0.6902
S023:TRUE and S019:TRUE	-7.0289	10.4042	0.5007
S023:TRUE and S014:TRUE	1.8000	5.5339	0.7456
S023:TRUE and S025:TRUE	10.1250	8.6398	0.2438
S023:TRUE and S021:TRUE	-0.6872	20.2490	0.9730
S023:TRUE and S001:TRUE	-14.9000	9.1769	0.1073
S023:TRUE and S022:TRUE	0.0250	4.5885	0.9957
S023:TRUE and S018:TRUE	-28.0206	20.0908	0.1659
S019:TRUE and S014:TRUE	0.0000	5.5339	1.0000
S019:TRUE and S025:TRUE	0.3794	7.6056	0.9603
S019:TRUE and S021:TRUE	2.9544	9.9015	0.7660
S019:TRUE and S022:TRUE	-4.7794	8.3264	0.5671
S019:TRUE and S018:TRUE	13.1044	9.7063	0.1798
S014:TRUE and S025:TRUE	-4.2794	3.9670	0.2831
S014:TRUE and S022:TRUE	-5.8500	4.5184	0.1981

TABLE E.3: Coefficients for
Overall_aggregate $\sim \sum \text{Course} \times \sum \text{Course}$ (Cont.)

Coefficients	Estimate	Std. Error	P-value
Intercept	13.3111	1.7591	0.0000
S026:TRUE	0.3660	0.8742	0.6760
S008:TRUE	2.3150	4.3337	0.5940
S007:TRUE	0.0385	0.7800	0.9607
S002:TRUE	0.6602	0.7261	0.3646
S023:TRUE	-0.5857	0.8172	0.4746
S019:TRUE	0.9072	0.7006	0.1972
S014:TRUE	1.1449	0.8221	0.1657
S025:TRUE	0.6369	0.9219	0.4906
S021:TRUE	-0.7244	0.9115	0.4279
S009:TRUE	3.0226	0.9354	0.0015
S001:TRUE	0.1173	0.9478	0.9017
S022:TRUE	1.4742	1.0128	0.1475
S018:TRUE	-1.4348	1.2829	0.2650
S006:TRUE	-0.4189	1.3476	0.7563
S016:TRUE	0.9985	0.9247	0.2818
SProj.30Cr:TRUE	-2.6541	4.4778	0.5542
SProj.20Cr:TRUE	2.7979	1.0319	0.0074
S005:TRUE	0.9908	0.9338	0.2903
S010:TRUE	-0.1191	3.1807	0.9702
S024:TRUE	0.1588	3.1935	0.9604
S013:TRUE	0.3207	3.8022	0.9329
S012:TRUE	-0.5667	2.2688	0.8031
S017:TRUE	-6.2272	2.1592	0.0045
S011:TRUE	-3.3192	1.5137	0.0298
S027:TRUE	1.9838	1.4115	0.1618
S028:TRUE	-0.7670	1.2277	0.5330
S029:TRUE	-0.0848	1.5791	0.9572

TABLE E.4: Coefficients for
Level4_aggregate $\sim \sum$ Course

Coefficients	Estimate	Std. Error	P-value
Intercept	8.4778	3.9154	0.0325
S026:TRUE	135.7416	146.9625	0.3577
S008:TRUE	-145.9069	135.5612	0.2841
S007:TRUE	-32.2206	48.4493	0.5074
S002:TRUE	63.8817	49.5752	0.2002
S023:TRUE	6.5222	4.4879	0.1490
S019:TRUE	0.6556	14.5847	0.9642
S014:TRUE	5.0250	9.2715	0.5889
S025:TRUE	-11.9778	24.1719	0.6212
S021:TRUE	4.4972	25.1761	0.8586
S009:TRUE	55.5989	49.9572	0.2682
S001:TRUE	7.7944	8.7036	0.3725
S022:TRUE	9.2472	10.0329	0.3587
S018:TRUE	22.7111	45.3587	0.6176
S006:TRUE	-107.1397	141.5189	0.4506
S016:TRUE	0.5000	4.0043	0.9009
SProj.30Cr:TRUE	124.0328	157.0210	0.4313
SProj.20Cr:TRUE	-3.6250	24.8010	0.8841
S005:TRUE	1.5222	2.7043	0.5747
S010:TRUE	36.8069	65.4189	0.5748
S024:TRUE	-92.0713	103.4349	0.3753
S013:TRUE	-81.6866	104.1859	0.4347
S012:TRUE	0.3111	4.4879	0.9449
S017:TRUE	-1.5000	4.0043	0.7087
S011:TRUE	0.6056	5.8798	0.9182
S027:TRUE	-87.3213	103.6004	0.4011
S028:TRUE	-89.2033	103.5822	0.3910
S029:TRUE	-90.0713	103.5259	0.3862
S026:TRUE and S008:TRUE	123.4947	112.5910	0.2751
S026:TRUE and S007:TRUE	106.4559	97.7263	0.2784
S026:TRUE and S002:TRUE	-54.4918	89.1924	0.5425
S026:TRUE and S023:TRUE	-67.5889	55.0140	0.2219
S026:TRUE and S019:TRUE	-61.3876	89.6361	0.4949
S026:TRUE and S014:TRUE	-68.9598	95.5334	0.4719
S026:TRUE and S025:TRUE	-53.3515	78.0424	0.4957
S026:TRUE and S021:TRUE	12.0736	49.2983	0.8070
S026:TRUE and S009:TRUE	5.0972	45.8141	0.9116
S026:TRUE and S001:TRUE	-137.2168	137.3865	0.3201
S026:TRUE and S022:TRUE	208.0353	258.9366	0.4235
S026:TRUE and S018:TRUE	-144.7988	202.1462	0.4753
S026:TRUE and S016:TRUE	7.4304	118.6895	0.9502

TABLE E.5: Coefficients for
Level4_aggregate $\sim \sum \text{Course} \times \sum \text{Course}$

Coefficients	Estimate	Std. Error	P-value
S026:TRUE and SProj.20Cr:TRUE	-110.5599	117.6633	0.3495
S026:TRUE and S005:TRUE	-163.5099	205.1795	0.4272
S026:TRUE and S012:TRUE	-187.3298	302.6182	0.5372
S026:TRUE and S017:TRUE	34.7286	154.3950	0.8224
S008:TRUE and S019:TRUE	4.2006	37.9983	0.9122
S007:TRUE and S002:TRUE	-30.3403	29.3470	0.3035
S007:TRUE and S023:TRUE	21.1345	51.2339	0.6808
S007:TRUE and S019:TRUE	-16.9261	30.8225	0.5840
S007:TRUE and S014:TRUE	-24.5066	28.0164	0.3836
S007:TRUE and S025:TRUE	24.6594	28.1682	0.3832
S007:TRUE and S021:TRUE	-21.1315	18.4657	0.2550
S007:TRUE and S001:TRUE	34.0556	15.1564	0.0266
S007:TRUE and S022:TRUE	37.5733	34.1764	0.2740
S002:TRUE and S023:TRUE	-36.2060	23.5645	0.1273
S002:TRUE and S019:TRUE	20.3014	16.2483	0.2142
S002:TRUE and S014:TRUE	-13.7407	12.9336	0.2904
S002:TRUE and S025:TRUE	-22.2532	23.0108	0.3356
S002:TRUE and S021:TRUE	19.5444	20.9296	0.3524
S002:TRUE and S001:TRUE	-48.8435	31.9913	0.1297
S002:TRUE and S022:TRUE	-31.1310	20.8656	0.1386
S002:TRUE and S018:TRUE	-14.1694	26.7637	0.5976
S002:TRUE and S016:TRUE	7.4292	10.7037	0.4891
S002:TRUE and SProj.20Cr:TRUE	-3.8282	23.8687	0.8729
S023:TRUE and S019:TRUE	-2.9667	11.4221	0.7956
S023:TRUE and S014:TRUE	1.6056	7.1699	0.8232
S023:TRUE and S025:TRUE	13.0000	27.1901	0.6335
S023:TRUE and S021:TRUE	-8.1417	24.1586	0.7368
S023:TRUE and S001:TRUE	-9.8556	9.2239	0.2876
S023:TRUE and S022:TRUE	-1.8361	6.6175	0.7819
S023:TRUE and S018:TRUE	-34.9944	42.0265	0.4068
S023:TRUE and S016:TRUE	-1.5611	4.7483	0.7430
S023:TRUE and SProj.20Cr:TRUE	7.6417	22.1081	0.7303
S019:TRUE and S014:TRUE	1.1667	5.6629	0.8372
S019:TRUE and S025:TRUE	2.6806	7.7829	0.7312
S019:TRUE and S021:TRUE	3.3889	10.1323	0.7387
S019:TRUE and S022:TRUE	-5.7639	8.5205	0.5002
S019:TRUE and S018:TRUE	11.5556	9.9325	0.2472
S014:TRUE and S025:TRUE	-4.0972	4.0595	0.3151
S014:TRUE and S022:TRUE	-6.5944	4.6237	0.1566

TABLE E.6: Coefficients for
Level4_aggregate $\sim \sum \text{Course} \times \sum \text{Course}$ (Cont.)

Coefficients	Estimate	Std. Error	P-value
Intercept	-5.4937	18.1288	0.7630
S026:TRUE	2.8503	1.9298	0.1455
S008:TRUE	-3.7271	3.7063	0.3191
S007:TRUE	2.6887	1.9086	0.1647
S002:TRUE	4.7447	3.9048	0.2296
S023:TRUE	3.0236	1.9549	0.1278
S019:TRUE	5.1568	3.9923	0.2020
S014:TRUE	2.4604	3.0499	0.4234
S025:TRUE	3.7222	3.5985	0.3056
S021:TRUE	3.9365	2.9013	0.1805
S009:TRUE	3.4021	1.9352	0.0844
S001:TRUE	4.1422	2.6463	0.1234
S022:TRUE	3.4414	3.0427	0.2630
S018:TRUE	6.3769	4.9723	0.2052
S006:TRUE	3.2114	2.4241	0.1908
S016:TRUE	3.9092	3.1706	0.2229
SProj.20Cr:TRUE	5.6085	5.3644	0.3004
S005:TRUE	7.7368	5.5787	0.1712
S024:TRUE	5.4944	6.5216	0.4032
S027:TRUE	3.2222	3.1778	0.3151
S028:TRUE	0.8722	3.2880	0.7918
S029:TRUE	1.7339	3.8244	0.6521

TABLE E.7: Coefficients for
Overall_aggregate $\sim \sum$ Course
(first class students)

Coefficients	Estimate	Std. Error	P-value
Intercept	-26.5018	43.7761	0.5477
S026:TRUE	6.0156	6.4696	0.3570
S008:TRUE	-9.0022	10.6313	0.4012
S007:TRUE	5.9956	5.5822	0.2881
S002:TRUE	12.1689	10.4686	0.2507
S023:TRUE	3.4622	3.3698	0.3093
S019:TRUE	11.3356	10.4686	0.2842
S014:TRUE	5.3156	7.1252	0.4592
S025:TRUE	6.5773	7.3871	0.3776
S021:TRUE	4.9622	4.8207	0.3084
S009:TRUE	5.2067	5.3094	0.3316
S001:TRUE	8.1156	6.9136	0.2461
S022:TRUE	6.2956	6.7835	0.3579
S018:TRUE	10.2205	9.1803	0.2710
S006:TRUE	6.4705	6.7141	0.3399
S016:TRUE	8.2156	8.1008	0.3155
SProj.20Cr:TRUE	9.6395	11.9210	0.4226
S005:TRUE	15.1356	13.2269	0.2581
S024:TRUE	11.2046	14.4438	0.4416
S027:TRUE	6.0773	7.1832	0.4016
S028:TRUE	3.7273	7.2347	0.6087
S029:TRUE	4.5891	7.5045	0.5437
S026:TRUE and S008:TRUE	2.4200	4.0545	0.5533
S026:TRUE and S007:TRUE	-1.5000	2.4376	0.5412
S026:TRUE and S002:TRUE	-3.5489	4.3680	0.4205
S026:TRUE and S019:TRUE	-2.3200	4.1985	0.5831
S002:TRUE and S019:TRUE	-3.7533	2.5950	0.1544

TABLE E.8: Coefficients for
Overall_aggregate $\sim \sum \text{Course} \times \sum \text{Course}$
(first class students)

Coefficients	Estimate	Std. Error	P-value
Intercept	-8.7807	17.5259	0.6184
S026:TRUE	2.7182	1.8656	0.1509
S008:TRUE	-4.5712	3.5830	0.2075
S007:TRUE	2.8530	1.8452	0.1279
S002:TRUE	6.0957	3.7749	0.1122
S023:TRUE	3.9457	1.8899	0.0415
S019:TRUE	6.2125	3.8596	0.1133
S014:TRUE	2.7273	2.9484	0.3591
S025:TRUE	4.3131	3.4788	0.2204
S021:TRUE	5.2831	2.8048	0.0650
S009:TRUE	3.7507	1.8708	0.0500
S001:TRUE	3.9922	2.5583	0.1245
S022:TRUE	4.0689	2.9415	0.1723
S018:TRUE	6.1527	4.8070	0.2060
S006:TRUE	3.7761	2.3434	0.1129
S016:TRUE	4.2397	3.0651	0.1723
SProj.20Cr:TRUE	5.4837	5.1860	0.2950
S005:TRUE	9.2250	5.3932	0.0929
S024:TRUE	5.7511	6.3047	0.3657
S027:TRUE	2.7297	3.0721	0.3782
S028:TRUE	2.1881	3.1787	0.4942
S029:TRUE	2.4405	3.6972	0.5120

TABLE E.9: Coefficients for
Level4_aggregate $\sim \sum$ Course
(first class students)

Coefficients	Estimate	Std. Error	P-value
Intercept	-48.2476	41.9182	0.2553
S026:TRUE	8.8399	6.1951	0.1599
S008:TRUE	-16.0227	10.1801	0.1219
S007:TRUE	9.9066	5.3453	0.0699
S002:TRUE	16.4399	10.0243	0.1074
S023:TRUE	5.9066	3.2267	0.0733
S019:TRUE	15.9955	10.0243	0.1170
S014:TRUE	8.8399	6.8228	0.2012
S025:TRUE	10.4257	7.0736	0.1469
S021:TRUE	8.4066	4.6161	0.0747
S009:TRUE	7.9828	5.0841	0.1228
S001:TRUE	10.8399	6.6202	0.1080
S022:TRUE	10.2399	6.4956	0.1214
S018:TRUE	12.9764	8.7907	0.1463
S006:TRUE	9.4254	6.4292	0.1490
S016:TRUE	11.8399	7.7570	0.1334
SProj.20Cr:TRUE	15.4990	11.4151	0.1808
S005:TRUE	20.8288	12.6655	0.1065
S024:TRUE	17.9764	13.8308	0.1998
S027:TRUE	8.8424	6.8784	0.2046
S028:TRUE	8.3007	6.9277	0.2366
S029:TRUE	8.5532	7.1860	0.2397
S026:TRUE and S008:TRUE	5.3389	3.8824	0.1753
S026:TRUE and S007:TRUE	-2.8333	2.3341	0.2306
S026:TRUE and S002:TRUE	-4.9034	4.1826	0.2467
S026:TRUE and S019:TRUE	-4.3222	4.0203	0.2876
S002:TRUE and S019:TRUE	-3.6000	2.4849	0.1538

TABLE E.10: Coefficients for
Level4_aggregate $\sim \sum \text{Course} \times \sum \text{Course}$
(first class students)

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